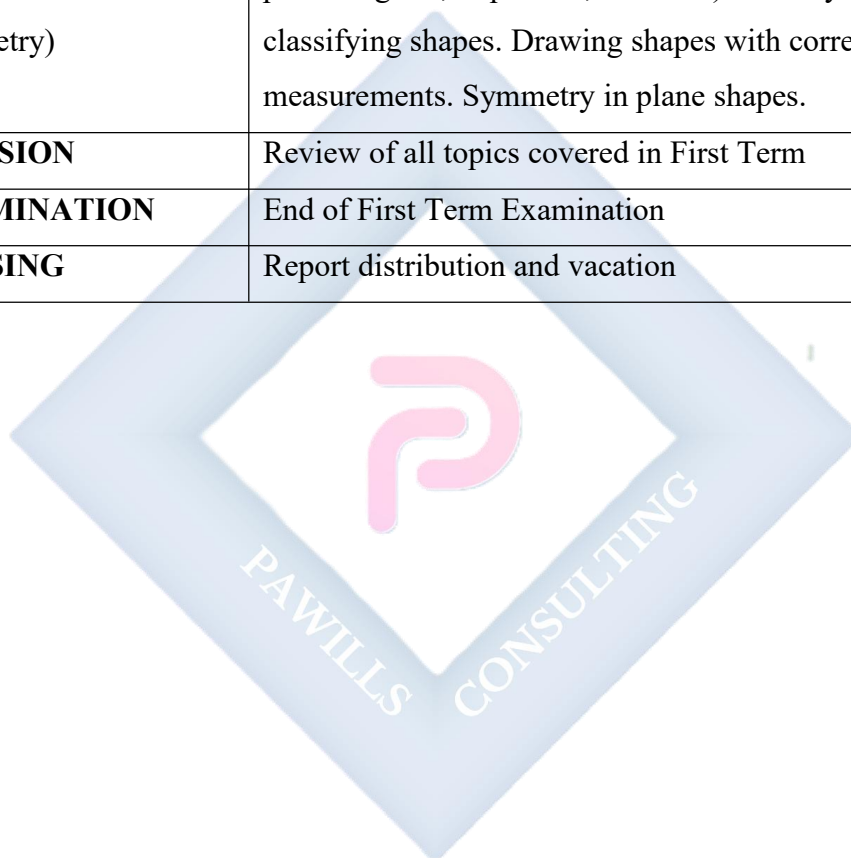


PRIMARY 5 MATHEMATICS SCHEME OF WORK

FIRST TERM

Week	Topic	Content
1	Whole Numbers I (Numbers & Numeration)	Place value up to millions (reading, writing, and identifying place values). Comparing and ordering whole numbers up to 7 digits. Roman numerals (I-1000). Number patterns and sequences.
2	Addition and Subtraction I (Basic Operations)	Addition of whole numbers up to 5-6 digits without and with carrying. Subtraction of whole numbers up to 5-6 digits without and with borrowing. Word problems involving addition and subtraction. Estimation and approximation.
3	Use of Number Line (Basic Operations)	Representing numbers on a number line. Addition using number line (moving forward). Subtraction using number line (moving backward). Understanding positive and negative movements on number line.
4	Money I (Mensuration & Geometry)	Nigerian currency: Naira and Kobo conversion. Addition and subtraction of money in decimal notation. Shopping calculations and giving change. Profit and loss (simple problems).
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Length (Mensuration & Geometry)	Standard units of length (mm, cm, m, km). Converting between units of length. Measuring lengths using rulers and tape measures. Perimeter of rectangles, squares, and triangles. Word problems involving length.
8	Weight (Mensuration & Geometry)	Standard units of weight/mass (g, kg, tonne). Converting between units of weight. Weighing objects using scales.

		Addition and subtraction of weights. Word problems involving weight.
9	Time I (Mensuration & Geometry)	Reading time on analog and digital clocks (12-hour and 24-hour). Converting between 12-hour and 24-hour formats. Calculating duration and elapsed time. Calendar work: days, weeks, months, years.
10	Plane Shapes I (Mensuration & Geometry)	Properties of 2D shapes (triangle, square, rectangle, parallelogram, trapezium, rhombus). Identifying and classifying shapes. Drawing shapes with correct measurements. Symmetry in plane shapes.
11	REVISION	Review of all topics covered in First Term
12	EXAMINATION	End of First Term Examination
13	CLOSING	Report distribution and vacation



PRIMARY 5 MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: WHOLE NUMBERS I (NUMBERS & NUMERATION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Read and write whole numbers up to millions.
2. Identify the place value of digits in whole numbers.
3. Compare and order numbers up to 7 digits using symbols ($<$, $>$, $=$).
4. Read and write Roman numerals up to 1000 (M).

CONTENT

1. Place Value up to Millions

Think of place value as a house where every digit lives. Each room in the house has a different name and value, getting bigger as you move to the left. We read large numbers by grouping them in threes from the right, using commas to separate the groups.

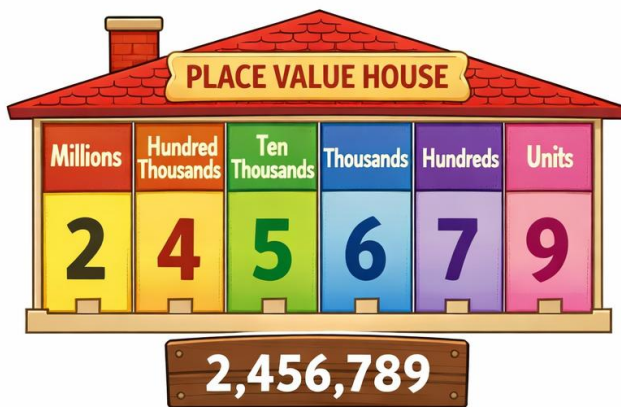
○ The Order (From Left to Right):

- Millions (M)
- Hundred Thousands (HTh)
- Ten Thousands (TTh)
- Thousands (Th)
- Hundreds (H)
- Tens (T)
- Units (U) or Ones.

○ Example: Let's explore the number **2,456,789**.

- The digit **2** lives in the **Millions** place. It means 2 million (2,000,000).
- The digit **4** lives in the **Hundred Thousands** place. It means 400,000.

- The digit **5** lives in the **Ten Thousands** place. It means 50,000.
- The digit **6** lives in the **Thousands** place. It means 6,000.
- The digit **7** lives in the **Hundreds** place. It means 700.
- The digit **8** lives in the **Tens** place. It means 80.
- The digit **9** lives in the **Units** place. It means 9.



2. Comparing and Ordering Whole Numbers

To compare two numbers, we start from the leftmost digit (the biggest place value house) and move to the right until we find a difference. It's like deciding which of two buildings is taller by comparing floor by floor from the top.

○ Symbols:

- $>$ means **Greater than** (The wider open side faces the bigger number).
- $<$ means **Less than** (The pointed end faces the smaller number).
- $=$ means **Equal to** (Both sides are the same).

- **Example:** Compare 54,000 and 56,000.
 - Both have 5 ten-thousands.
 - Move right: Compare the thousands place. 4 is less than 6.
 - Therefore, $54,000 < 56,000$. We say "Fifty-four thousand is less than fifty-six thousand."

2. Roman Numerals (I - 1000)

Long ago, the Romans used letters from their alphabet instead of the numbers we use today. We still see them on clock faces, in book chapters, or for special dates.

- **Basic Symbols:**
 - $I = 1$ (Like one finger)
 - $V = 5$ (Think of one hand with all five fingers)
 - $X = 10$ (Two V's put together, or two hands)
 - $L = 50$
 - $C = 100$ (Think of "Century" which is 100 years)
 - $D = 500$
 - $M = 1000$ (Think of "Millennium" which is 1000 years)
- **The Two Golden Rules:**
 - i. **If a smaller numeral is to the RIGHT of a bigger one, you ADD.**
 - Example: $VI = 5 (V) + 1 (I) = 6$. It's like having 5 naira and getting 1 more naira.
 - ii. **If a smaller numeral is to the LEFT of a bigger one, you SUBTRACT.**
 - Example: $IV = 5 (V) - 1 (I) = 4$. It's like needing 5 naira but you only have 1, so you are 1 *away from* 5.
 - Numerals are usually written from largest to smallest value. We don't write the same symbol more than three times in a row. For 4, we write IV, not IIII.

3. Number Patterns

A number pattern is like a secret code or a rhythm in numbers. Once you find the rule, you can predict what comes next.

- **Example Pattern:** 2, 4, 6, 8, ____
 - Look at the jump from 2 to 4 (+2), 4 to 6 (+2), 6 to 8 (+2).
 - **The Rule is: "Add 2."**
 - The next number is $8 + 2 = 10$.

EVALUATION

1. In the number 8,361,045, which digit is in the hundred thousands place? What is its value?
2. Write the number "Three million, seventy thousand and fifteen" in figures.
3. Use $<$, $>$, or $=$ to fill the gap: 707,707 ____ 770,077.
4. Write the number 49 in Roman Numerals.
5. Find the rule and complete the pattern: 5, 10, 15, 20, ____.

ASSIGNMENT

1. **(Oral)** Tell your parent or guardian your home address. Ask them to help you identify any numbers in it that are greater than 10,000.
2. **(Written)** Arrange these numbers from the smallest to the biggest: 8,888; 8,088; 8,808.
3. **(Practical/Drawing)** Draw a simple clock face. Instead of numbers 1 to 12, write the correct Roman numerals at each position.

WEEK 2: ADDITION AND SUBTRACTION I (BASIC OPERATIONS)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Add whole numbers up to 6 digits involving carrying.
2. Subtract whole numbers up to 6 digits involving borrowing (renaming).
3. Estimate answers by rounding off numbers.

4. Solve word problems involving addition and subtraction.

CONTENT

1. Addition of Whole Numbers

Adding is like combining groups of items. For large numbers, we stack them vertically to make sure digits with the same place value are in the same column (Units under Units, Tens under Tens).

- **Carrying (Regrouping):** When the sum of a column is 10 or more, that value is too big for its "house." We write down the last digit (the "ones" part) in that column and **carry** the first digit (the "tens" part) over to the next column on the left. Think of it as moving a bundle of ten sticks to the next place value house.
- **Example:** $24,500 + 13,600$.
 - Step 1: Arrange vertically.
 - Step 2: Add Units: $0+0=0$. Add Tens: $0+0=0$.
 - Step 3: Add Hundreds: $5+6=11$. Write down **1** in the hundreds column, **carry 1** (which is 10 hundreds, or 1 thousand) to the thousands column.
 - Step 4: Add Thousands: $4+3=7$, plus the carried 1 = 8.
 - Step 5: Add Ten Thousands: $2+1=3$.
 - Answer: **38,100**.

$$\begin{array}{r}
 24,500 \\
 + 13,600 \\
 \hline
 5,600 \\
 8,00 \\
 \hline
 38,100
 \end{array}$$

The diagram illustrates the addition of 24,500 and 13,600. A red '1' is shown above the thousands column with a dashed arrow pointing to the hundreds column, indicating a carry. Vertical dashed lines align the digits in each column. The intermediate sums 5,600 and 8,00 are shown in blue, and the final result 38,100 is shown in red.

Subtraction of Whole Numbers

Subtraction is like taking away or finding the difference. We also stack numbers vertically. Sometimes, the top digit in a column is smaller than the bottom digit. We cannot take a bigger number from a smaller one, so we need to **borrow** (or rename).

- **Borrowing (Renaming):** We "borrow" 1 from the neighbor to the left. Remember, that 1 is worth 10 in the current column. The column we borrowed from decreases by 1.
- **Example:** $542 - 368$.
 - Step 1: Arrange vertically.
 - Step 2: Units: 2 is smaller than 8. Borrow 1 ten from the tens column. The 4 in the tens becomes 3. The 2 in the units becomes 12. Now, $12 - 8 = 4$.
 - Step 3: Tens: We now have 3 (after borrowing). 3 is smaller than 6. Borrow 1 hundred from the hundreds column. The 5 becomes 4. The 3 becomes 13. $13 - 6 = 7$.

- Step 4: Hundreds: We now have 4. $4 - 3 = 1$.
- Answer: **174**.

2. Estimation and Approximation

Estimation is making a smart, quick guess close to the real answer. It's useful for checking if our final answer is reasonable. We do this by **rounding off** numbers to the nearest ten, hundred, or thousand before calculating.

- **Rounding Rule:** Look at the digit to the right of the place you are rounding to. If it is 5 or more, round up. If it is 4 or less, round down.
- **Example:** Estimate $49 + 52$.
 - Round 49 to the nearest ten: $49 \rightarrow 50$ (because 9 is 5 or more).
 - Round 52 to the nearest ten: $52 \rightarrow 50$ (because 2 is 4 or less).
 - Estimated sum: $50 + 50 = 100$. (The actual sum is 101, so our estimate is close!).

3. Word Problems

Read the problem carefully. Look for keywords that tell you which operation to use.

- **Addition keywords:** total, sum, altogether, combined, in all, both.
- **Subtraction keywords:** difference, remain, left, how many more, how much less, decrease.

EVALUATION

1. Add: $305,172 + 189,456$.
2. Subtract: $70,001 - 45,678$.
3. Estimate the sum by rounding to the nearest hundred: $487 + 324$.
4. The population of Village A is 12,480 and Village B is 15,920. What is their combined population?
5. Tunde had ₦5,000. He bought a school bag for ₦3,250. How much money does he have left?

ASSIGNMENT

1. **(Written)** Calculate: $800,005 - 456,789$. Show your borrowing steps clearly.
2. **(Oral)** With a friend, one person says a 4-digit number. The other person quickly rounds it to the nearest thousand and explains why.
3. **(Practical)** Go shopping with an adult (real or pretend). Pick two items with prices. Estimate the total cost by rounding, then calculate the exact amount to see how close your estimate was.

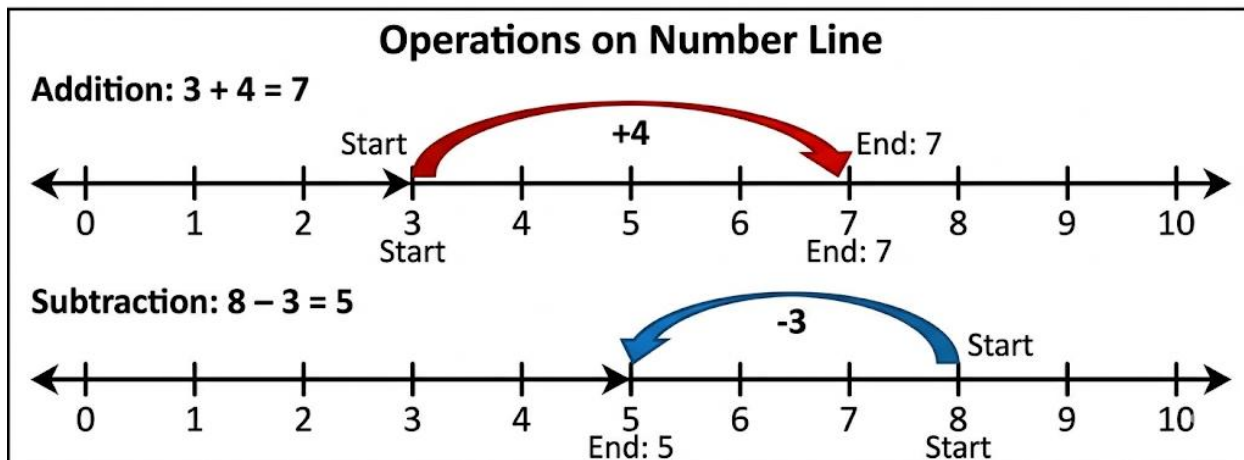


OBJECTIVES

1. Draw a number line representing positive numbers.
2. Perform addition using a number line (moving forward).
3. Perform subtraction using a number line (moving backward).
4. Demonstrate positive and negative movements on a number line.

1. Representing Numbers on a Number Line

- We usually start at **0**.
- As we move to the **RIGHT**, numbers get **bigger**. This is the positive direction.
- As we move to the **LEFT** from zero, numbers get **smaller**. This can lead to negative numbers (which we will learn more about later).



2. Addition using Number Line

Adding means combining quantities. On a number line, we show this by starting at the first number and making jumps to the **RIGHT**.

- **Example:** $3 + 4$.

- Step 1: Start at **3**.
- Step 2: Since we are adding **4**, make 4 jumps to the right.
- Step 3: Each jump moves 1 step: $3 \rightarrow 4$ (1st jump), $4 \rightarrow 5$ (2nd), $5 \rightarrow 6$ (3rd), $6 \rightarrow 7$ (4th).
- You land on **7**. So, $3 + 4 = 7$.

3. Subtraction using Number Line

Subtracting means taking away. On a number line, we show this by starting at the first number (the minuend) and making jumps to the **LEFT** (backwards).

- **Example:** $8 - 3$.

- Step 1: Start at **8**.
- Step 2: Since we are subtracting **3**, make 3 jumps to the left.
- Step 3: Each jump moves 1 step: $8 \rightarrow 7$ (1st jump), $7 \rightarrow 6$ (2nd), $6 \rightarrow 5$ (3rd).
- You land on **5**. So, $8 - 3 = 5$.

4. Understanding Movements

We can think of the number line like a road. Moving forward (right) is like walking towards the market (increasing). Moving backward (left) is like walking back home (decreasing). If you go left past your house (0), you might go to your neighbor's house (negative numbers).

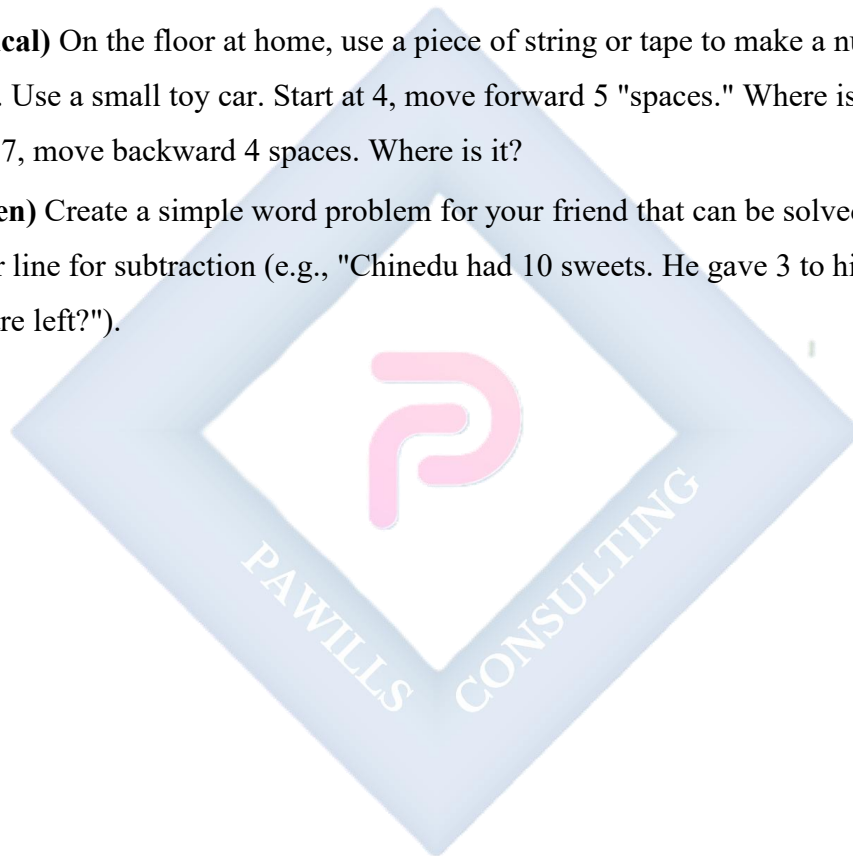
EVALUATION

1. Draw a number line from 0 to 12.
2. Using your number line, show and solve $5 + 6$.

3. Using your number line, show and solve $11 - 4$.
4. If you start at 2 on a number line and move 7 steps to the right, what number do you reach?
5. If you start at 9 and move 5 steps to the left, what number do you reach? What operation did you just perform?

ASSIGNMENT

1. **(Drawing)** Draw a number line to solve $9 + 8 = ?$. Color your jumps with a crayon.
2. **(Practical)** On the floor at home, use a piece of string or tape to make a number line from 0 to 10. Use a small toy car. Start at 4, move forward 5 "spaces." Where is the car? Now start at 7, move backward 4 spaces. Where is it?
3. **(Written)** Create a simple word problem for your friend that can be solved using the number line for subtraction (e.g., "Chinedu had 10 sweets. He gave 3 to his sister. How many are left?").



WEEK 4: MONEY I (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Convert Naira to Kobo and Kobo to Naira.
2. Add and subtract money correctly using decimal notation.
3. Calculate change in shopping transactions.
4. Solve simple profit and loss problems.

CONTENT

1. Nigerian Currency Conversion

Our money is called Naira (₦) and Kobo (k). They are related, just like metres and centimetres.

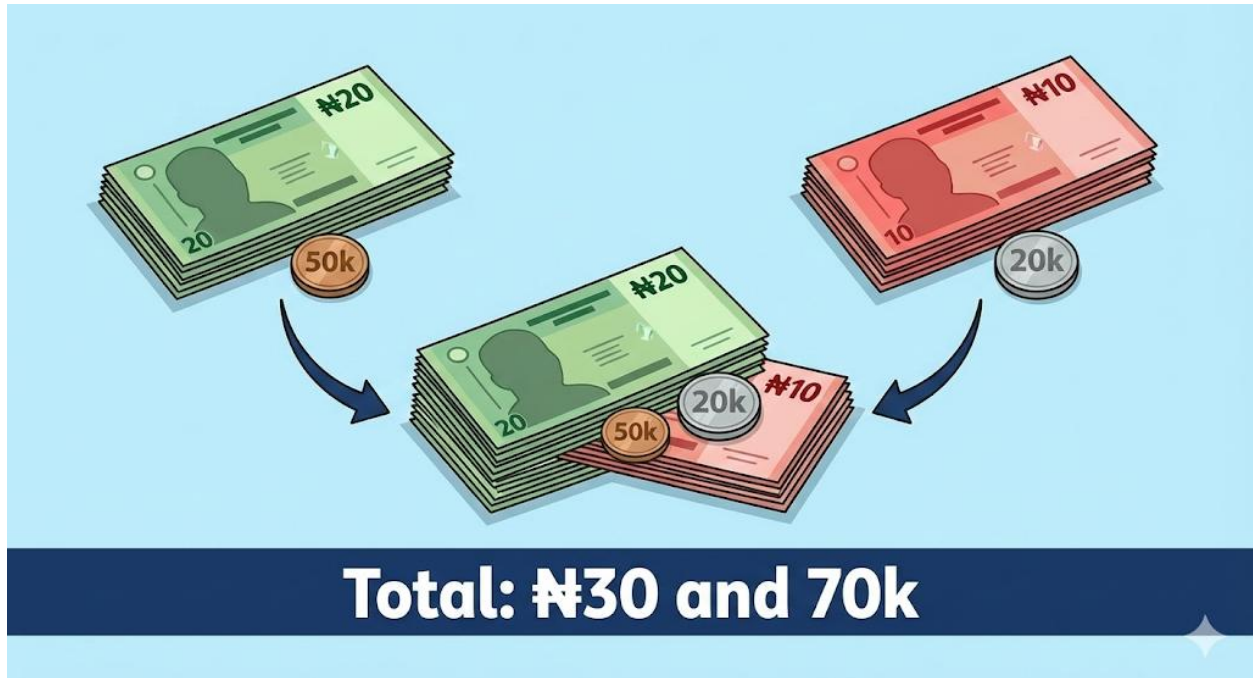
- **100 Kobo = 1 Naira.** This is the most important rule!
- **To change Naira to Kobo:** Multiply by 100. This is like changing a ₦1 note into 100 kobo coins.
 - Example: $₦5 = 5 \times 100 = 500\text{k}$.
- **To change Kobo to Naira:** Divide by 100. This is like gathering 100 kobo coins to get a ₦1 note.
 - Example: $350\text{k} = 350 \div 100 = ₦3.50$. We write this as three Naira, fifty Kobo. The dot (.) separates Naira from Kobo.

2. Addition and Subtraction of Money

Always write money amounts with the decimal point in line. Think of the dot as a separator between the "Naira house" and the "Kobo house."

- **Example:** $₦20.50 + ₦10.20$.
 - Step 1: Arrange vertically, aligning the decimal points.
 - Step 2: Add the Kobo: $50\text{k} + 20\text{k} = 70\text{k}$. Write .70.
 - Step 3: Add the Naira: $₦20 + ₦10 = ₦30$.

- Answer: **₦30.70**.
- Subtraction follows the same rule with aligning dots. Remember to borrow if needed, knowing that 1 Naira = 100 Kobo.



Shopping and Giving Change

This is a very important life skill.

- **Total Cost:** Add up the prices of all items you are buying.
- **Change:** Money you get back = Money you gave to seller (Amount Paid) – Total Cost.
- **Example:** A textbook costs ₦850. You give the seller ₦1,000.
 - $\text{Change} = \text{₦}1,000 - \text{₦}850 = \text{₦}150$.

3. Profit and Loss

This is about buying and selling in business.

- **Cost Price (CP):** The price a trader buys an item.
- **Selling Price (SP):** The price a trader sells the item.
- **Profit:** Occurs when SP is **greater than** CP. $\text{Profit} = \text{SP} - \text{CP}$. (Trader is happy!)
- **Loss:** Occurs when SP is **less than** CP. $\text{Loss} = \text{CP} - \text{SP}$. (Trader is sad.)

EVALUATION

1. Convert ₦12.75 to Kobo.
2. Add ₦275.30 and ₦150.85.
3. A packet of biscuits costs ₦220. If you pay with a ₦500 note, how much change should you receive?
4. Mrs. Bello bought a basket for ₦800 and sold it for ₦1,200. Calculate her profit.
5. A trader bought shoes for ₦3,000 but had to sell them for ₦2,500. Did he make a profit or a loss? Calculate how much.

ASSIGNMENT

1. **(Oral)** Ask a parent the price of three common food items at home (e.g., milk, bread, eggs). Practice saying the total cost if you bought one of each.
2. **(Written)** A trader bought 5 meters of cloth at ₦500 per meter. She sold it all for ₦3,000. Calculate her total profit or loss.
3. **(Practical/Drawing)** Draw a simple shop scene. Draw two items with price tags (e.g., a book for ₦250, a pen for ₦150). Draw a ₦500 note. Calculate and write the total cost and the change you would get.

WEEK 5: MIDTERM EXAMINATION

(Assessment covering Weeks 1-4)



WEEK 6: MIDTERM BREAK

(Holiday)



WEEK 7: LENGTH (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. List standard units of length (mm, cm, m, km).
2. Convert between units of length (e.g., cm to m).
3. Measure lengths of objects using a ruler.
4. Calculate the perimeter of rectangles and squares.

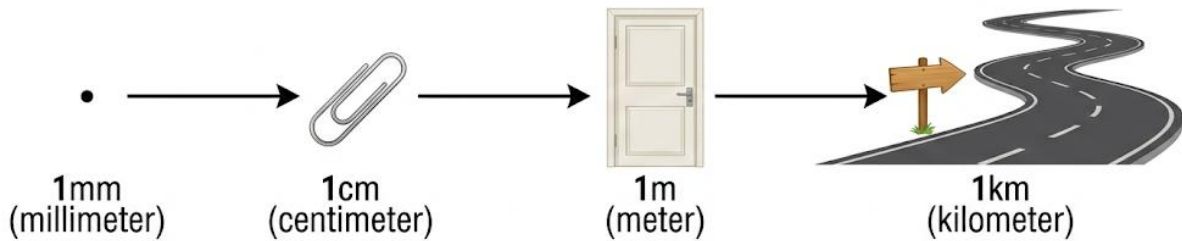
CONTENT

1. Standard Units of Length

We measure length to know how long, short, tall, or wide something is. We use different units for different tasks, just like using a cup for water and a bucket for mopping.

- **Millimetre (mm):** This is very, very small. It's about the thickness of your pupil ID card or a grain of sand. We use mm for measuring tiny things.
- **Centimetre (cm):** This is small. It's about the width of your fingernail or a paperclip. Your ruler is 30cm long. We use cm a lot in the classroom.
- **Metre (m):** This is bigger. It's about the height of a door or the length of a big stride. A teacher's metre rule is 1m long. We use m for measuring rooms, fences, and people.
- **Kilometre (km):** This is very long. It's used for distances between places. The distance from your school to the main market might be a few kilometres.

UNITS OF LENGTH SCALE

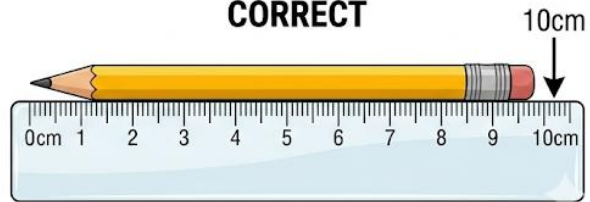


CORRECT VS INCORRECT MEASURING

INCORRECT



CORRECT



2. Conversion Between Units

We can change from one unit to another because they are connected by 10s, 100s, and 1000s. Remember this ladder:

- 10 millimetres (mm) = 1 centimetre (cm)
- 100 centimetres (cm) = 1 metre (m)
- 1000 metres (m) = 1 kilometre (km)
- **The Golden Rule:** To change from a **bigger unit** to a **smaller unit**, you **MULTIPLY** (you get more of the small units). To change from a **smaller unit** to a **bigger unit**, you **DIVIDE** (you get fewer of the big units).
- **Example 1 (Big to Small):** Convert 2m to cm.
 - Since 1m = 100cm, we multiply: $2 \times 100 = 200\text{cm}$.
- **Example 2 (Small to Big):** Convert 5000m to km.
 - Since 1000m = 1km, we divide: $5000 \div 1000 = 5\text{km}$.

3. Measuring Lengths Correctly

When using a ruler, you must be careful to start from the correct point to get an accurate measurement.

- **Step 1:** Place the object you are measuring along the ruler, starting at the "**0**" **mark**. Do not start from the very end of the ruler.
- **Step 2:** Look straight down at the ruler to avoid a "parallax error" (seeing the measurement from the side, which can be wrong).
- **Step 3:** Read the number on the ruler where the object ends.

4. Perimeter

Perimeter is the total distance **around** a shape. Think of it as the length of a fence you would need to surround a garden.

- **Rectangle:** A rectangle has two lengths and two breadths (widths).
 - **Formula:** Perimeter = Length + Breadth + Length + Breadth
 - A quicker way: Perimeter = $2 \times (\text{Length} + \text{Breadth})$. This is because there are two of each side.
 - **Example:** A rectangle is 10cm long and 4cm wide.
 - Perimeter = $2 \times (10\text{cm} + 4\text{cm}) = 2 \times 14\text{cm} = \mathbf{28\text{cm}}$.
- **Square:** A square has all four sides equal.
 - **Formula:** Perimeter = Side + Side + Side + Side
 - A quicker way: Perimeter = $4 \times \text{Length of one side}$.
 - **Example:** A square has a side of 5cm.
 - Perimeter = $4 \times 5\text{cm} = \mathbf{20\text{cm}}$.

EVALUATION

1. Which unit (mm, cm, m, km) is best for measuring: a) The length of your school field? b) The thickness of your maths textbook?
2. Convert 7 metres into centimetres.
3. Use your ruler to measure the length of your pen in centimetres.
4. Find the perimeter of a square whose side is 8cm.

5. A rectangular picture frame is 25cm long and 15cm wide. What is its perimeter?

ASSIGNMENT

1. **(Practical)** At home, use a measuring tape (or a string and ruler) to measure and record the length and width of your bed in metres (m) and centimetres (cm).
2. **(Written)** Convert the following: a) 5km to metres. b) 250cm to metres.
3. **(Drawing)** Draw a rectangle with a length of 6cm and a width of 3cm on a squared sheet. Calculate and write its perimeter around the drawing.



WEEK 8: WEIGHT (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. State standard units of weight (g, kg).
2. Convert grams to kilograms and vice versa.
3. Read weighing scales correctly.
4. Add and subtract weights in kg and g.

CONTENT

1. Standard Units of Weight

Weight tells us how heavy or light an object is. We use scales to measure it.

- **Gram (g):** Used for light objects. A sachet of sugar, a pencil, or a key might weigh a few grams.
- **Kilogram (kg):** Used for heavier objects. A bag of rice, a bunch of plantains, or a small dog is measured in kilograms. 1 kilogram feels quite heavy in your hand.
- **Tonne (t):** Used for very heavy things like cars, lorries, or elephants. (1 tonne = 1000 kg).

2. Conversion Between Grams and Kilograms

This is similar to Naira and Kobo!

- **1000 grams (g) = 1 kilogram (kg).**
- **To change kg to g:** Multiply by 1000. (You are breaking the big kg into many small g).
 - Example: $3\text{kg} = 3 \times 1000 = 3000\text{g}$.
- **To change g to kg:** Divide by 1000. (You are packing the small g into big kg).
 - Example: $4500\text{g} = 4500 \div 1000 = 4.5\text{kg}$ or **4kg 500g**.



3. Reading Weighing Scales

Scales can be digital (showing numbers) or analog (with a needle). For analog scales:

- **Step 1:** Look at where the needle is pointing.
- **Step 2:** Check what number each mark on the scale represents (e.g., does each small line mean 100g or 200g?).
- **Step 3:** Read the number carefully. If the needle is between two numbers, estimate the weight.

4. Addition and Subtraction of Weight

We add and subtract weights just like numbers, but we must keep the units (kg and g) in separate columns. Sometimes we need to regroup.

- **Example (Addition):** Add 3kg 450g and 2kg 600g.
 - Step 1: Add grams: $450\text{g} + 600\text{g} = 1050\text{g}$.
 - Step 2: Since 1050g is more than 1000g, we convert 1000g to 1kg. So, $1050\text{g} = 1\text{kg } 50\text{g}$.
 - Step 3: Add kilograms: $3\text{kg} + 2\text{kg} + 1\text{kg (from the grams)} = 6\text{kg}$.
 - Step 4: Final answer: **6kg 50g**.

- **Subtraction** follows similar steps, sometimes requiring us to "borrow" 1kg (which is 1000g) from the kg column.

EVALUATION

1. Would you measure the weight of a chicken in grams or kilograms? Why?
2. Convert 6.25kg to grams.
3. Look at the diagram of a scale.



What weight is shown?

4. Add: $4\text{kg } 750\text{g} + 1\text{kg } 300\text{g}$.
5. Subtract: $5\text{kg } 200\text{g} - 2\text{kg } 850\text{g}$. (Hint: You will need to borrow).

ASSIGNMENT

1. **(Oral)** At home, find three items: one you think weighs less than 100g, one about 1kg, and one more than 5kg. Tell an adult your estimates.
2. **(Written)** A baby weighed 3.8kg at birth. After one month, she weighs 4.5kg. How much weight has she gained (in kg and g)?
3. **(Practical)** Help an adult in the kitchen. Look at the packaging of two different food items (e.g., flour, rice). Write down their weights in grams and kilograms. Practice adding them together.

WEEK 9: TIME I (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Read time on analog and digital clocks.
2. Convert time between 12-hour and 24-hour systems.
3. Calculate duration between two times.
4. Recall the number of days in different months.

CONTENT

1. Reading Time

- **Analog Clock (with hands):**

- **Short hand:** Hour hand. It moves slowly. When it points exactly to a number, it is that hour.
- **Long hand:** Minute hand. It moves faster. Each number it points to represents 5 minutes (1=5min, 2=10min, etc.).
- **Example:** If the short hand is between 2 and 3, and the long hand is on the 6, the time is **2:30**.

- **Digital Clock:** Shows numbers directly, like **14:30** or **2:30**.



2. 12-hour and 24-hour Clock Systems

- **12-hour clock:** Uses numbers 1 to 12, followed by **a.m.** (midnight to noon) or **p.m.** (noon to midnight). We use this when speaking.
- **24-hour clock:** Uses numbers 00:00 to 23:59. No "a.m./p.m." is used. It is common on timetables, phones, and digital devices.
- **Conversion Rules:**
 - **From 12-hour (a.m.) to 24-hour:** Write the hour with two digits. 8:00 a.m. becomes **08:00**.
 - **From 12-hour (p.m.) to 24-hour:** Add 12 to the hour. 3:00 p.m. becomes $(3+12) = \mathbf{15:00}$. 12:00 p.m. (noon) is **12:00**.
 - **From 24-hour to 12-hour:** For hours 00:00-11:59, it's a.m. (01:00 = 1:00 a.m.). For hours 12:00-23:59, subtract 12 to get the p.m. time (17:00 = 5:00 p.m.).

3. Calculating Duration

Duration is how long an event lasts. To find it, we find the difference between the **End Time** and the **Start Time**.

- **Example:** Assembly starts at 8:15 a.m. and ends at 8:45 a.m.
 - Method 1: Count forward in minutes. From 8:15 to 8:45 is **30 minutes**.
 - Method 2: Subtract: $8:45 - 8:15 = \mathbf{0 \text{ hours } 30 \text{ minutes}}$.

4. Calendar and Time Relationships

It's important to know how time units relate.

- 60 seconds = 1 minute
- 60 minutes = 1 hour
- 24 hours = 1 day
- **Days in Months:** Use the rhyme or your knuckles!
 - Make two fists. Starting from the knuckle of your index finger as January, the valley between knuckles as February, next knuckle as March, etc. Every knuckle month has 31 days, every valley month has 30 (except February).

EVALUATION

1. Draw an analog clock showing the time 10 past 4 (4:10).
2. Write 11:20 p.m. in 24-hour format.
3. Write 06:45 in 12-hour format (a.m./p.m.).
4. School closes at 2:00 p.m. If there are 6 periods of 40 minutes each with a 30-minute break, what time does school start? (Calculate the total learning time first).
5. How many days are in the month of August?

ASSIGNMENT

1. **(Practical)** For one day, keep a simple log. Write down the 24-hour time when you do these: wake up, eat lunch, go to bed.
2. **(Written)** A movie starts at 15:30 and lasts for 1 hour 45 minutes. At what time does it end? Give your answer in both 24-hour and 12-hour formats.
3. **(Oral)** Recite the "days in the month" rhyme to a family member. Ask them to test you on a random month.

WEEK 10: PLANE SHAPES I (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify common 2D shapes (triangle, square, rectangle, rhombus, trapezium, parallelogram).
2. List properties of these shapes (sides, corners/angles, parallel sides).
3. Draw plane shapes with specific measurements.
4. Identify lines of symmetry in plane shapes.

CONTENT

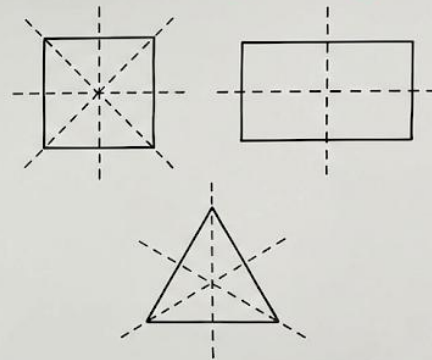
1. Properties of 2D (Plane) Shapes

Plane shapes are flat shapes that have only two dimensions: length and breadth. They have sides (edges) and corners (vertices/angles).

- **Triangle:** Has **3** sides and **3** corners. The sum of its inside angles is always 180° .
- **Square:** Has **4 equal** sides and **4** right angles (90° corners). Opposite sides are parallel.
- **Rectangle:** Has **4** sides. **Opposite sides are equal and parallel.** It has **4** right angles.
- **Parallelogram:** Has **4** sides. **Opposite sides are equal and parallel.** It looks like a slanted rectangle. Opposite angles are equal.
- **Rhombus:** Has **4 equal** sides, like a square. Opposite sides are parallel. It looks like a pushed-over square or a kite. Opposite angles are equal.
- **Trapezium (US: Trapezoid):** Has **4** sides, but **only one pair of sides is parallel.**

2D Shapes Properties Table		
Shape	Drawing	Properties
		3 sides, 3 angles
		4 equal sides, 4 right angles
		4 sides, opposite sides equal, 4 right angles
		4 sides, opposite sides parallel & equal
		4 equal sides, opposite angles equal
		4 sides, one pair of parallel sides

Lines of Symmetry



2. Drawing Shapes Accurately

To draw shapes well, you need the right tools and careful steps.

- **Use a ruler** for straight lines. Make sure corners meet exactly.
- **Use a protractor** to measure angles if you need to be precise (e.g., for a right angle).
- **Example - Drawing a rectangle 5cm by 3cm:**
 - i. Draw one side 5cm long.
 - ii. At one end, draw a perpendicular (90°) line 3cm long.
 - iii. At the other end of the first line, draw another 3cm line.
 - iv. Join the ends of the two 3cm lines with a 5cm line.

3. Symmetry

A line of symmetry is an imaginary line where you can fold a shape, and both halves match perfectly. It's like a mirror line.

- **Square:** Has **4** lines of symmetry (vertical, horizontal, two diagonals).
- **Rectangle:** Has **2** lines of symmetry (vertical and horizontal through the center).
- **Equilateral Triangle:** Has **3** lines of symmetry.

- **Isosceles Triangle:** Has **1** line of symmetry.
- **Circle:** Has infinite lines of symmetry (any line through the center).

EVALUATION

1. Name a shape that has four sides but only one pair of parallel sides.
2. What is the key difference between a square and a rhombus?
3. Draw a parallelogram with a pair of sides 4cm long and another pair 2cm long. (Label your sides).
4. How many lines of symmetry does a rectangle have? Draw them on a rectangle.
5. Look at the shape of a standard traffic warning sign (a triangle). What type of triangle is it usually, and how many lines of symmetry does it have?

ASSIGNMENT

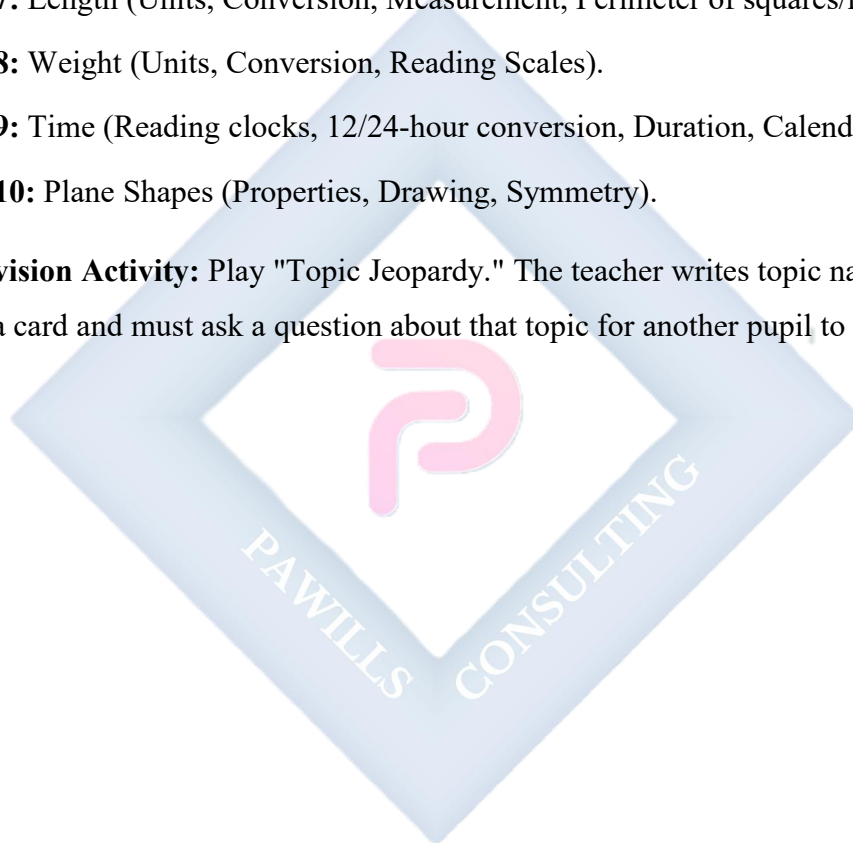
1. **(Drawing)** On a squared paper, draw: a) A square of side 5 squares. b) A rectangle that is 6 squares long and 4 squares wide.
2. **(Practical/Hunt)** At home or in school, go on a "shape hunt." Find and list one real-life object for each: Triangle, Rectangle, Square, Circle.
3. **(Written)** A shape has 4 equal sides, but its angles are not 90 degrees. Its opposite sides are parallel. What is its name? How many lines of symmetry does it have?

WEEK 11: REVISION

Review of all topics covered from Weeks 1-10.

- **Week 1:** Place Value (Millions), Comparing Numbers, Roman Numerals.
- **Week 2:** Addition & Subtraction with carrying/borrowing, Estimation, Word Problems.
- **Week 3:** Using the Number Line for addition and subtraction.
- **Week 4:** Money (Conversion, Addition/Subtraction, Profit/Loss, Change).
- **Week 7:** Length (Units, Conversion, Measurement, Perimeter of squares/rectangles).
- **Week 8:** Weight (Units, Conversion, Reading Scales).
- **Week 9:** Time (Reading clocks, 12/24-hour conversion, Duration, Calendar).
- **Week 10:** Plane Shapes (Properties, Drawing, Symmetry).

Suggested Revision Activity: Play "Topic Jeopardy." The teacher writes topic names on cards. A pupil picks a card and must ask a question about that topic for another pupil to answer.



PRIMARY 5 MATHEMATICS SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	Fractions I (Numbers & Numeration)	Understanding fractions: numerator and denominator. Types of fractions (proper, improper, mixed numbers). Equivalent fractions. Comparing and ordering fractions. Converting between improper fractions and mixed numbers.
2	Fractions II (Numbers & Numeration)	Addition of fractions with same denominators. Subtraction of fractions with same denominators. Introduction to addition and subtraction of fractions with different denominators (simple cases). Word problems involving fractions.
3	Multiplication I (Numbers & Numeration)	Multiplication tables (revision up to 12×12). Multiplication of 2-digit by 2-digit numbers. Multiplication of 3-digit by 1-digit and 2-digit numbers. Properties of multiplication (commutative, associative, distributive). Word problems involving multiplication.
4	Division I (Numbers & Numeration)	Division as inverse of multiplication. Division of 2-digit and 3-digit numbers by 1-digit numbers. Division with remainders. Checking division using multiplication. Word problems involving division.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Temperature (Mensuration & Geometry)	Understanding temperature and thermometers. Reading temperatures in degrees Celsius ($^{\circ}\text{C}$). Comparing temperatures (hotter, colder, warmer, cooler). Temperature in daily life (weather, body temperature, cooking). Addition and subtraction of temperatures.
8	Area I (Mensuration)	Understanding area as the space covered by a surface.

	& Geometry)	Standard units of area (cm^2 , m^2 , km^2). Finding area of rectangles and squares ($\text{length} \times \text{width}$). Finding area of triangles ($\frac{1}{2} \times \text{base} \times \text{height}$). Word problems involving area.
9	Volume I (Mensuration & Geometry)	Understanding volume as space occupied by an object. Standard units of volume (cm^3 , m^3). Finding volume of cubes and cuboids ($\text{length} \times \text{width} \times \text{height}$). Estimating and comparing volumes.
10	Capacity (Mensuration & Geometry)	Understanding capacity as the amount a container can hold. Standard units of capacity (ml, cl, l). Converting between units of capacity. Measuring capacity using measuring cylinders and containers. Word problems involving capacity.
11	REVISION	Review of all topics covered in Second Term
12	EXAMINATION	End of Second Term Examination
13	CLOSING	Report distribution and vacation

PRIMARY 5 MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: FRACTIONS I (NUMBERS & NUMERATION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

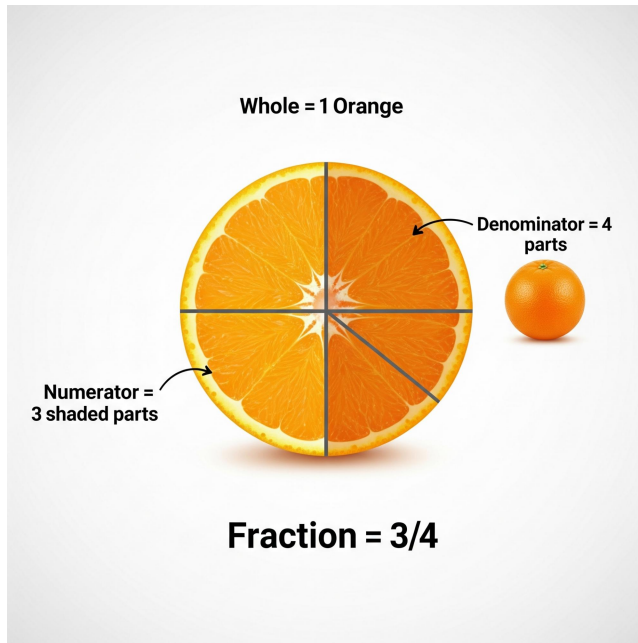
1. Define a fraction and identify the numerator and denominator.
2. Differentiate between proper, improper, and mixed fractions.
3. Generate equivalent fractions for a given number.
4. Convert improper fractions to mixed numbers and vice versa.

CONTENT

1. Understanding Fractions

A fraction represents a part of a whole. Imagine you have a whole orange. If you cut it into equal parts, each part is a fraction of the whole orange. The key is that the parts must be equal.

- **Parts of a Fraction:**
 - **Denominator (Bottom Number):** Tells us **into how many equal parts** the whole is divided. Think: "Down" for denominator.
 - **Numerator (Top Number):** Tells us **how many of those parts** we are talking about or have.
- **Example:** In the fraction $\frac{3}{4}$,
 - The denominator **4** means the whole is divided into **4 equal parts**.
 - The numerator **3** means we are talking about **3 of those parts**.
 - So, $\frac{3}{4}$ means "three out of four equal parts."



2. Types of Fractions

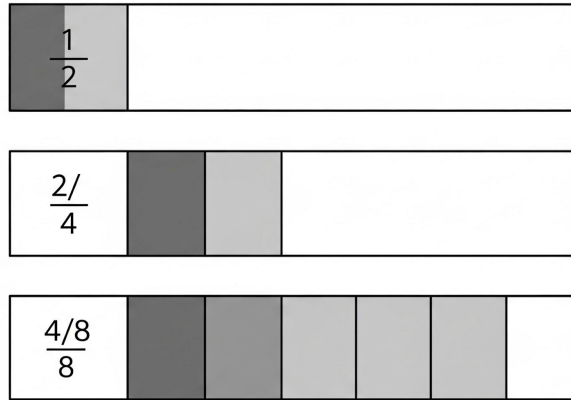
- **Proper Fraction:** The numerator is **smaller than** the denominator. Its value is **less than 1 whole**. Example: $\frac{2}{5}$ (You have 2 parts out of 5, which is less than a full thing).
- **Improper Fraction:** The numerator is **equal to or larger than** the denominator. Its value is **1 or more**. Example: $\frac{5}{2}$ (You have 5 halves; that's more than one whole). $\frac{4}{4}$ is also improper because it equals exactly 1 whole.
- **Mixed Number:** This is a mix of a **whole number** and a **proper fraction**. It shows clearly how many wholes and extra parts there are. Example: $2\frac{1}{2}$ (This means 2 whole oranges and half of another one).

3. Equivalent Fractions

These are different fractions that represent the **same amount or value**. You can create them by multiplying or dividing the numerator and denominator by the same number.

- **Why it works:** It's like cutting the same pizza into more slices. $\frac{1}{2}$ of a pizza is the same amount as $\frac{2}{4}$ of the same pizza, you just have more, smaller slices.
- **Example:** $\frac{1}{2}$ is equivalent to $\frac{2}{4}$, $\frac{3}{6}$, and $\frac{50}{100}$.
- **How to find them:**

- **Multiply** top and bottom by the same number: $\frac{1}{2} \times (2/2) = 2/4$.
- **Divide** top and bottom by the same number: $4/8 \div (4/4) = \frac{1}{2}$.



4. Conversions

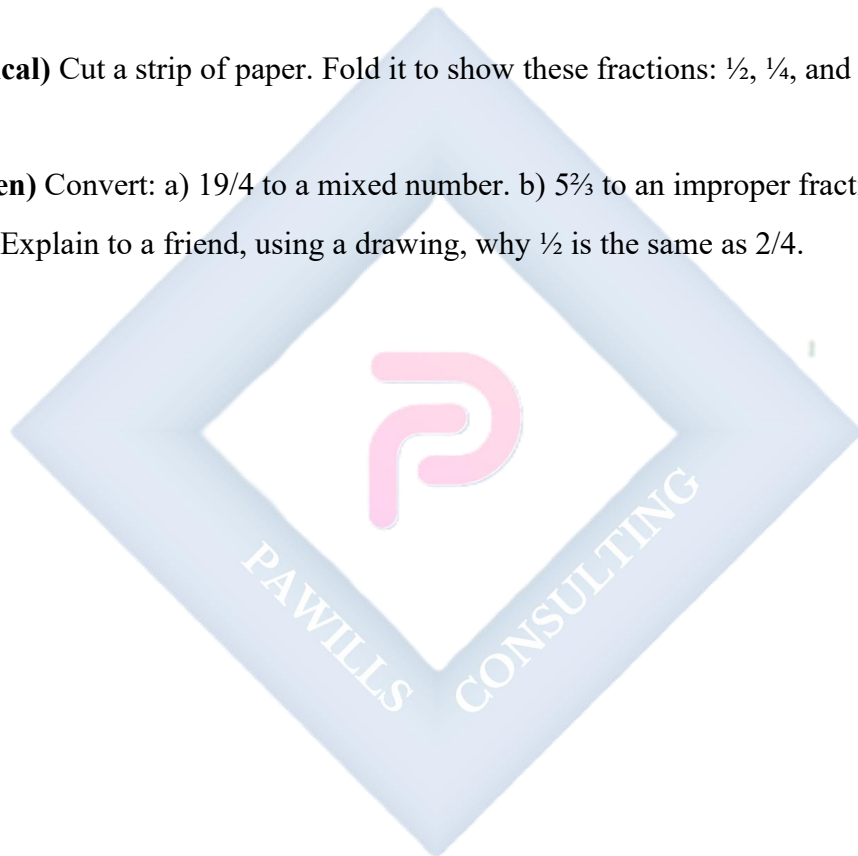
- **Improper Fraction to Mixed Number:**
 - Divide** the numerator by the denominator.
 - The **quotient** (answer) becomes the **whole number**.
 - The **remainder** becomes the new **numerator** over the original denominator.
 - **Example:** Convert $11/4$.
 - $11 \div 4 = 2$ remainder 3.
 - So, $11/4 = 2\frac{3}{4}$.
- **Mixed Number to Improper Fraction:**
 - Multiply** the whole number by the denominator.
 - Add** the result to the numerator.
 - Place that answer over the original denominator.
 - **Example:** Convert $3\frac{1}{2}$.
 - $(3 \times 2) + 1 = 6 + 1 = 7$.
 - So, $3\frac{1}{2} = 7/2$.

EVALUATION

1. In the fraction $\frac{7}{12}$, what does the number 12 tell you?
2. Classify these as proper, improper, or mixed: a) $\frac{8}{3}$ b) $1\frac{2}{3}$ c) $\frac{4}{9}$.
3. Write two fractions that are equivalent to $\frac{2}{3}$.
4. Convert $\frac{17}{5}$ to a mixed number.
5. Convert $4\frac{3}{5}$ to an improper fraction.

ASSIGNMENT

1. **(Practical)** Cut a strip of paper. Fold it to show these fractions: $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{3}{8}$. Label each part.
2. **(Written)** Convert: a) $\frac{19}{4}$ to a mixed number. b) $5\frac{2}{3}$ to an improper fraction.
3. **(Oral)** Explain to a friend, using a drawing, why $\frac{1}{2}$ is the same as $\frac{2}{4}$.



WEEK 2: FRACTIONS II (NUMBERS & NUMERATION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Add and subtract fractions with the same denominators.
2. Add and subtract fractions with different denominators.
3. Solve simple word problems involving fractions.

CONTENT

1. Addition and Subtraction (Same Denominators)

This is simple! If the denominators are the same, it means the "size of the pieces" is the same. You just add or subtract the number of pieces (numerators). The denominator stays the same.

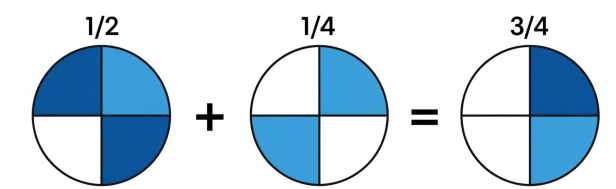
- **Rule:** Add/Subtract the numerators, keep the denominator.
- **Example (Addition):** $\frac{2}{7} + \frac{3}{7} = \frac{(2+3)}{7} = \frac{5}{7}$.
 - Think: "2 slices of a pizza cut into 7, plus 3 more slices of the same size pizza, equals 5 slices."
- **Example (Subtraction):** $\frac{6}{8} - \frac{1}{8} = \frac{(6-1)}{8} = \frac{5}{8}$.

2. Addition and Subtraction (Different Denominators)

If denominators are different, the "piece sizes" are different. You cannot add a half and a quarter directly. You must first make the denominators the same (find a **common denominator**).

- **Step 1:** Find the Lowest Common Multiple (LCM) of the denominators. This becomes your new, common denominator.
- **Step 2:** Change each fraction to an equivalent fraction with the new denominator.
- **Step 3:** Now add or subtract the numerators as in Step 1.
- **Example:** $\frac{1}{2} + \frac{1}{4}$.
 - Step 1: LCM of 2 and 4 is 4.

- Step 2: Change $\frac{1}{2}$ to $\frac{2}{4}$. Multiply top and bottom by 2: $(1 \times 2)/(2 \times 2) = \frac{2}{4}$.
 $\frac{1}{4}$ stays as $\frac{1}{4}$.
- Step 3: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$.



3. Word Problems

Read carefully. Identify what is being asked and what operation (addition or subtraction) the keywords suggest.

- **Example:** Kemi ate $\frac{1}{5}$ of a cake and Tunde ate $\frac{2}{5}$ of the same cake. What fraction of the cake did they eat together?
 - They ate parts of the **same** whole cake (same denominator).
 - "Together" means **add**.
 - Calculation: $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$.
 - Answer: They ate $\frac{3}{5}$ of the cake together.

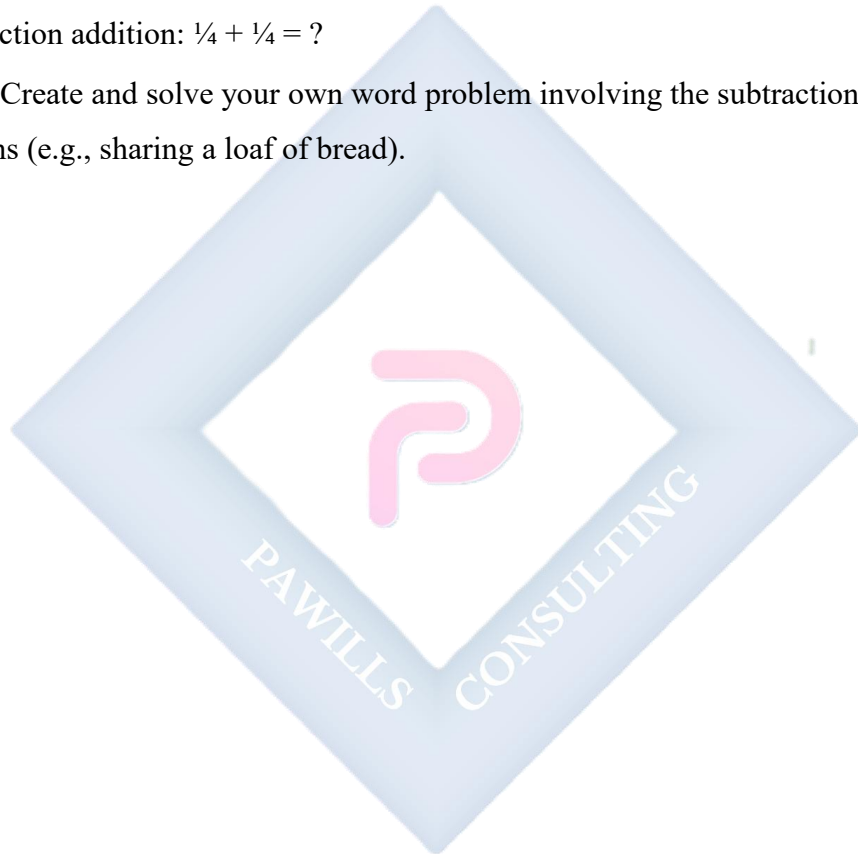
EVALUATION

1. Add: $\frac{5}{12} + \frac{1}{12}$.
2. Subtract: $\frac{7}{10} - \frac{3}{10}$.
3. Add: $\frac{1}{3} + \frac{1}{6}$. (Hint: Find a common denominator).

4. Subtract: $\frac{3}{4} - \frac{1}{2}$.
5. A water tank was $\frac{2}{3}$ full. After a storm added more water, it became $\frac{5}{6}$ full. What fraction of the tank was added by the storm?

ASSIGNMENT

1. **(Written)** Solve: a) $\frac{4}{5} + \frac{1}{10}$. b) $\frac{5}{6} - \frac{1}{3}$.
2. **(Practical)** Help an adult measure ingredients in the kitchen. If a recipe needs $\frac{1}{2}$ cup of oil and you only have a $\frac{1}{4}$ cup measure, how many times do you need to fill it? Write this as a fraction addition: $\frac{1}{4} + \frac{1}{4} = ?$
3. **(Oral)** Create and solve your own word problem involving the subtraction of two fractions (e.g., sharing a loaf of bread).



WEEK 3: MULTIPLICATION I (NUMBERS & NUMERATION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Multiply 2-digit numbers by 2-digit numbers.
2. Multiply 3-digit numbers by 1-digit and 2-digit numbers.
3. Identify properties of multiplication (commutative, associative, distributive).
4. Solve word problems involving multiplication.

CONTENT

1. Multiplication Review and 2-Digit by 2-Digit

Multiplication is repeated addition. The vertical method (long multiplication) helps organize our work.

- **Example:** 24×12 .
 - Think of it as 24 groups of 12, or $(24 \times 10) + (24 \times 2)$.
 - **Step 1:** Multiply 24 by the unit digit of 12 (2): $24 \times 2 = 48$.
 - **Step 2:** Multiply 24 by the tens digit of 12 (1, which means 10): $24 \times 10 = 240$. Write this shifted one place to the left (placing a zero in the units column helps).
 - **Step 3:** Add the two results: $48 + 240 = 288$.

2. Multiplication of 3-Digit Numbers

The same vertical method applies, with careful carrying over.

- **Example:** 123×5 .
 - Step 1: 5×3 (units) = 15. Write down 5, carry over 1 ten.
 - Step 2: 5×2 (tens) = 10, plus the carried 1 = 11. Write down 1, carry over 1 hundred.
 - Step 3: 5×1 (hundreds) = 5, plus the carried 1 = 6.

- Answer: **615**.

$$\begin{array}{r}
 123 \times 123 \\
 \underline{5} \\
 \hline
 5 \times 3 = 15 \\
 5 \times 2 = 10 + 11 \\
 5 \times 1 = 5 \quad 6 \\
 5 \times 1 = \text{hundreds} \\
 \hline
 615
 \end{array}$$

3. Properties of Multiplication

These are special rules that are always true and can make calculation easier.

- **Commutative Property:** You can swap the order of numbers. The product remains the same.
 - Example: $6 \times 4 = 24$ and $4 \times 6 = 24$.
- **Associative Property:** When multiplying three or more numbers, the way you group them doesn't change the product.
 - Example: $(2 \times 3) \times 4 = 6 \times 4 = 24$. $2 \times (3 \times 4) = 2 \times 12 = 24$.
- **Distributive Property:** Multiplying a number by a sum is the same as multiplying the number by each part of the sum and then adding.
 - Example: $4 \times (3 + 5) = (4 \times 3) + (4 \times 5) = 12 + 20 = 32$. This is useful for mental math: $6 \times 12 = 6 \times (10 + 2) = 60 + 12 = 72$.

4. Word Problems

Look for keywords like "total," "altogether," "product," "each," "times," "per," "for every."

- **Example:** If one exercise book costs ₦150, how much will 8 such books cost?
 - This is **8 groups of ₦150**.
 - Calculation: $150 \times 8 = \text{₦}1,200$.

EVALUATION

1. Calculate: 32×14 using the long multiplication method.
2. Calculate: 407×6 .
3. Which property is shown here? $9 \times 7 = 7 \times 9$.
4. Use the distributive property to solve 5×18 mentally. (Hint: $18 = 10 + 8$).
5. A crate holds 24 bottles of soft drink. How many bottles are in 15 crates?

ASSIGNMENT

1. **(Written)** Find the product of 56 and 23. Show your working.
2. **(Mental/Oral)** Use the distributive property to solve: 7×22 . Explain your steps to someone at home.
3. **(Practical)** Look at a pack of items at home (e.g., sachets of milk in a pack, eggs in a carton). Write down the number in one pack and calculate how many would be in 6 packs.

WEEK 4: DIVISION I (NUMBERS & NUMERATION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Explain division as the inverse (opposite) of multiplication.
2. Divide 2-digit and 3-digit numbers by 1-digit numbers.
3. Solve division problems with remainders.
4. Check division answers using multiplication.

CONTENT

1. Division as Inverse of Multiplication

Division is **sharing** or **grouping** equally. It undoes multiplication.

- Fact Family: If $4 \times 5 = 20$, then:
 - $20 \div 5 = 4$ (Sharing 20 into 5 equal groups gives 4 in each)
 - $20 \div 4 = 5$ (Sharing 20 into 4 equal groups gives 5 in each)

2. Long Division Method

This is a step-by-step way to divide larger numbers. We use the "DMSB" mnemonic: **D**ivide, **M**ultiply, **S**ubtract, **B**ring down.

- **Example:** $125 \div 5$.
 - **Step 1 (Divide):** 5 into the first digit (1) doesn't go. Take the first two digits: 5 into 12 goes **2** times. Write 2 above the 2.
 - **Step 2 (Multiply):** $2 \times 5 = 10$. Write 10 under the 12.
 - **Step 3 (Subtract):** $12 - 10 = 2$.
 - **Step 4 (Bring down):** Bring down the next digit (5) to make 25.
 - **Repeat:** 5 into 25 goes **5** times. Write 5 above the 5. Multiply ($5 \times 5 = 25$), subtract ($25 - 25 = 0$).
 - **Answer:** **25**.

$$\begin{array}{r}
 5 \overline{) 125} \\
 \underline{5} \\
 12 \\
 \underline{10} \\
 10 \\
 \underline{10} \\
 2 \\
 \text{Bring down the 5} \\
 25 \\
 \underline{25} \\
 25 \\
 \underline{25} \\
 0
 \end{array}$$

D. Divide $\rightarrow$ $12 \times 5 = 2 \text{ R } 2$
M. Multiply $\leftarrow$ $2 \times 5 = 10$
S. Subtract $\rightarrow$ $10 - 10 = 0$
S. Subtract $\rightarrow$ $25 \times 5 = 5$
B. Bring Down $\rightarrow$ 25
B. Bring Down $\rightarrow$ $5 \times 5 = 25$
 $25 - 25 = 0$

Answer: 25

3. Division with Remainders

Sometimes, numbers don't share perfectly. The amount left over is the **remainder**. It must be **less than** the divisor.

- **Example:** $30 \div 7$.
 - $7 \times 4 = 28$ (this is the closest to 30 without going over).
 - $30 - 28 = 2$.
 - So, $30 \div 7 = 4 \text{ remainder } 2$.

4. Checking Division

You can always check your answer using multiplication.

- **Check Formula:** $(\text{Quotient} \times \text{Divisor}) + \text{Remainder} = \text{Dividend}$.
- **Check for the example above:** $(4 \times 7) + 2 = 28 + 2 = 30$. Correct!

EVALUATION

1. If $6 \times 9 = 54$, write the two related division facts.
2. Solve: $84 \div 4$ using long division.
3. Solve: $57 \div 5$. State the quotient and remainder.
4. Check your answer for question 3 using the multiplication method.
5. 48 pencils are to be shared equally among 6 pupils. How many pencils does each get?

ASSIGNMENT

1. **(Written)** Calculate: a) $96 \div 6$ b) $153 \div 7$ (find the remainder).
2. **(Oral/Practical)** Get 17 small identical items (e.g., seeds, pebbles). Share them equally among 4 groups. How many are in each group? How many are left over? Write this as a division sentence.
3. **(Written)** A farmer harvested 100 mangoes. He packed them into bags of 8 mangoes each. How many full bags did he get? How many mangoes were left unpacked? (Find quotient and remainder).

WEEK 5: MIDTERM EXAMINATION

(Assessment covering Weeks 1-4)



WEEK 6: MIDTERM BREAK

(Holiday)



WEEK 7: TEMPERATURE (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

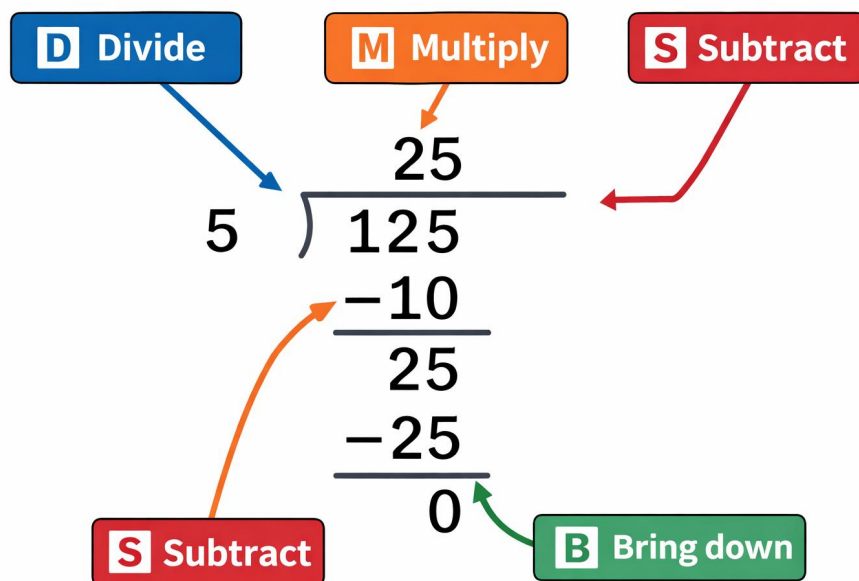
1. Define temperature and identify the instrument used to measure it.
2. Read temperatures in degrees Celsius ($^{\circ}\text{C}$).
3. Compare temperatures to determine which is hotter or colder.
4. Perform simple addition and subtraction of temperature values.

CONTENT

1. Understanding Temperature

Temperature is a measure of **how hot or cold** something is.

- **Instrument:** A **thermometer**. It has a bulb with liquid (often colored alcohol) that expands when hot and contracts when cold.
- **Unit:** We use **degrees Celsius ($^{\circ}\text{C}$)** in Nigeria and most of the world.



2. Reading and Comparing Temperatures

- **Key References:**

- **0°C:** Water freezes into ice. A very cold day.
- **20-25°C:** A typical comfortable room temperature in Nigeria.
- **37°C:** Normal human body temperature.
- **100°C:** Water boils. Very hot.

- **Comparison:** The **higher** the number in °C, the **hotter** it is. The **lower** the number, the **colder** it is.

- Example: 35°C is **hotter than** 25°C.

3. Temperature in Daily Life

- **Weather:** We check if it's a hot or cool day.
- **Health:** Doctors use thermometers to check for fever (body temperature above 37.5°C).
- **Cooking:** Recipes specify oven temperatures.

4. Calculating Temperature Changes

We can add or subtract temperature values just like ordinary numbers, but always include the unit (°C).

- **Example 1 (Increase):** If the morning temperature was 24°C and it rose by 7°C by afternoon, the new temperature is $24^{\circ}\text{C} + 7^{\circ}\text{C} = 31^{\circ}\text{C}$.
- **Example 2 (Decrease):** If the temperature drops from 30°C to 25°C, the change is $30^{\circ}\text{C} - 25^{\circ}\text{C} = \text{a drop of } 5^{\circ}\text{C}$.

EVALUATION

1. What is the normal human body temperature in °C?
2. Which is colder: 12°C or 18°C?
3. If a pot of soup is at 85°C and it cools down by 15°C, what is its new temperature?
4. What is the boiling point of water?
5. Look at the thermometer diagram.



What temperature is shown?

ASSIGNMENT

1. **(Practical/Drawing)** Draw a thermometer. Mark and label: 0°C , 25°C , 37°C , and 100°C .
2. **(Written)** The temperature in Lagos was 32°C . In Jos, it was 22°C . What is the difference in temperature between the two cities?
3. **(Oral)** Ask three family members what they think a "hot" day and a "cold" day feel like in $^{\circ}\text{C}$. Discuss why their answers might be different.

WEEK 8: AREA I (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define area.
2. State the standard units of area (cm^2 , m^2).
3. Calculate the area of rectangles and squares.
4. Calculate the area of triangles using the formula.

CONTENT

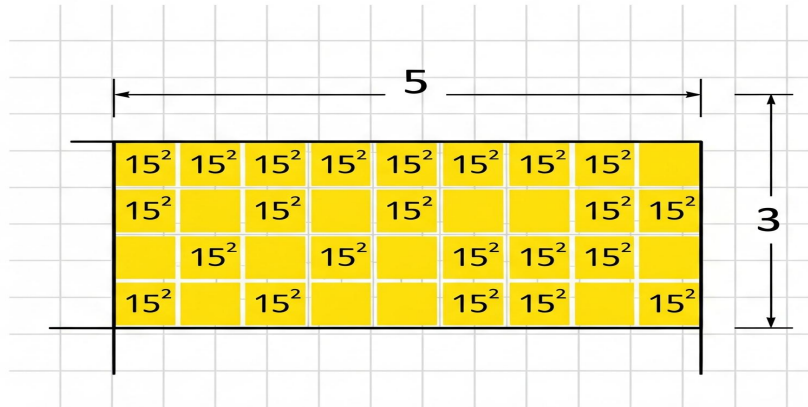
1. Understanding Area

Area is the amount of **flat surface** a shape covers. It is measured in **square units**.

- **Analogy:** Imagine covering your desk with square post-it notes. The number of notes needed is the area of the desk.
- **Units:**
 - **Square centimetre (cm^2):** For small areas (page of a book, a tile).
 - **Square metre (m^2):** For larger areas (classroom floor, a plot of land).

2. Area of Rectangles and Squares

- **Rectangle:** Area = **Length** $\times$ **Breadth (Width)**.
 - **Why?** If you multiply the number of squares along the length by the number along the width, you get the total number of squares.
 - **Example:** A rectangle 5cm long and 3cm wide.
 - Area = $5\text{cm} \times 3\text{cm} = 15 \text{ cm}^2$.
- **Square:** Since Length = Breadth, Area = **Side** $\times$ **Side** or **Side**².
 - **Example:** A square with sides of 4cm.
 - Area = $4\text{cm} \times 4\text{cm} = 16 \text{ cm}^2$.

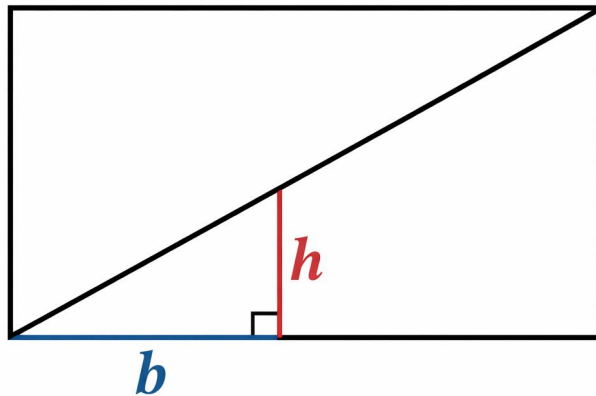


Length > Width = Area
 $5 \times 3 = 15$ square units

3. Area of Triangles

A triangle is always **half of a related rectangle or parallelogram**.

- **Formula:** Area = $\frac{1}{2} \times \text{Base} \times \text{Height}$ OR $(\text{Base} \times \text{Height}) \div 2$.
- **Important:** The **height** must be the **perpendicular height** (a straight line at 90°) from the base to the top vertex (corner), not just any side.
- **Example:** A triangle with a base of 6cm and a height of 4cm.
 - Area = $(6\text{cm} \times 4\text{cm}) \div 2 = 24\text{cm}^2 \div 2 = 12 \text{ cm}^2$.



$$A = \frac{1}{2} \times b \times h$$

EVALUATION

1. What is the difference between *perimeter* and *area*?
2. Calculate the area of a square with a side of 9cm.
3. A rectangular whiteboard is 2m long and 1m wide. What is its area in square metres (m²)?
4. Find the area of a triangle with a base of 10cm and a height of 5cm.
5. A rectangular garden has an area of 24m². If its length is 6m, what is its width?

ASSIGNMENT

1. **(Practical)** Use a ruler to measure the length and width of your math textbook cover in cm. Calculate its area.
2. **(Written)** A triangular flag has a base of 20cm and a height of 15cm. What is its area?
3. **(Drawing)** Draw a rectangle with an area of 12 cm² on squared paper. (Hint: What length and width can you use?).

WEEK 9: VOLUME I (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define volume.
2. State the standard units of volume (cm^3 , m^3).
3. Calculate the volume of cubes and cuboids.
4. Compare the volumes of different objects.

CONTENT

1. Understanding Volume

While **area** covers a flat surface (2D), **volume** measures the **space inside a solid object** (3D) – how much it can hold or the space it occupies.

- **Analogy:** The amount of water that fills a bottle is its volume (capacity). The amount of sand that fills a box is its volume.
- **Units: Cubic units** like cubic centimetres (cm^3) and cubic metres (m^3).

2. Volume of Cuboids (Rectangular Prisms)

A cuboid is a box shape (e.g., a chalk box, a book).

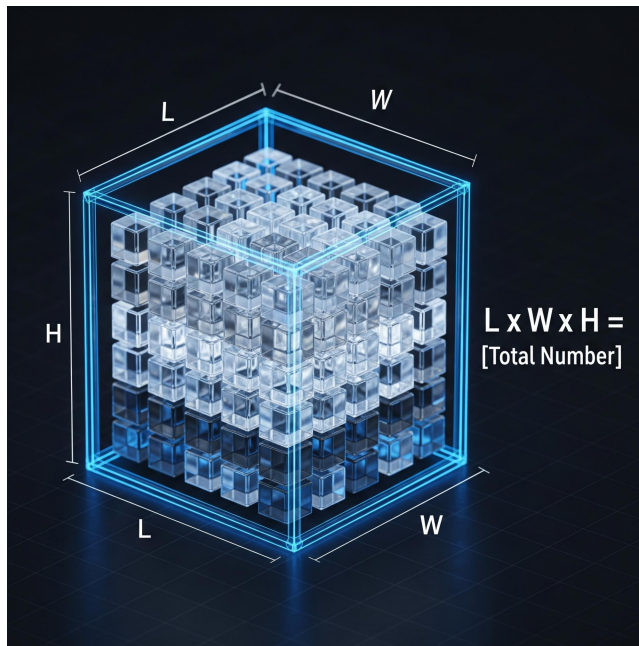
- **Formula:** Volume = **Length** $\times$ **Width** $\times$ **Height**.
- **Why?** Imagine filling the box with small 1cm^3 cubes. You can count how many cubes fit along the length, width, and height, and multiply these numbers.
- **Example:** A box is 4cm long, 3cm wide, and 2cm high.
 - Volume = $4\text{cm} \times 3\text{cm} \times 2\text{cm} = 24\text{ cm}^3$.

3. Volume of Cubes

A cube is a special cuboid where Length = Width = Height.

- **Formula:** Volume = **Side** $\times$ **Side** $\times$ **Side** or **Side**³.
- **Example:** A cube with sides of 3cm.

▪ $\text{Volume} = 3\text{cm} \times 3\text{cm} \times 3\text{cm} = 27 \text{ cm}^3.$



4. Comparing Volumes

The object with the larger cubic unit number has the larger volume.

- Example: A box of 150 cm^3 can hold more than a box of 120 cm^3 .

EVALUATION

1. Would you use cm^3 or m^3 to measure the volume of a classroom?
2. Calculate the volume of a cuboid with length=5cm, width=4cm, height=3cm.
3. What is the volume of a cube with a side length of 6cm?
4. Which container has a greater volume: Container A (L=10cm, W=5cm, H=2cm) or Container B (Side=6cm)? Show your working.
5. Explain the main difference between calculating *area* and *volume*.

ASSIGNMENT

1. **(Written)** Find the volume of a rectangular tank that is 8m long, 5m wide, and 2m deep.
2. **(Practical)** Find a small box at home (e.g., a matchbox, toothpaste box). Measure its length, width, and height in cm and calculate its approximate volume.

3. **(Drawing)** Draw a cube with a side of 4 cubes on isometric dotted paper (if available) or simply write its dimensions and calculated volume.



WEEK 10: CAPACITY (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define capacity.
2. List standard units of capacity (ml, cl, l).
3. Convert between Litres and Millilitres.
4. Solve simple problems involving capacity.

CONTENT

1. Understanding Capacity

Capacity is the **maximum amount of liquid** a container can hold. It is closely related to volume but is specifically used for fluids.

- **Analogy:** The volume of a water tank is the space inside it. The capacity is how many litres of water that space can hold.

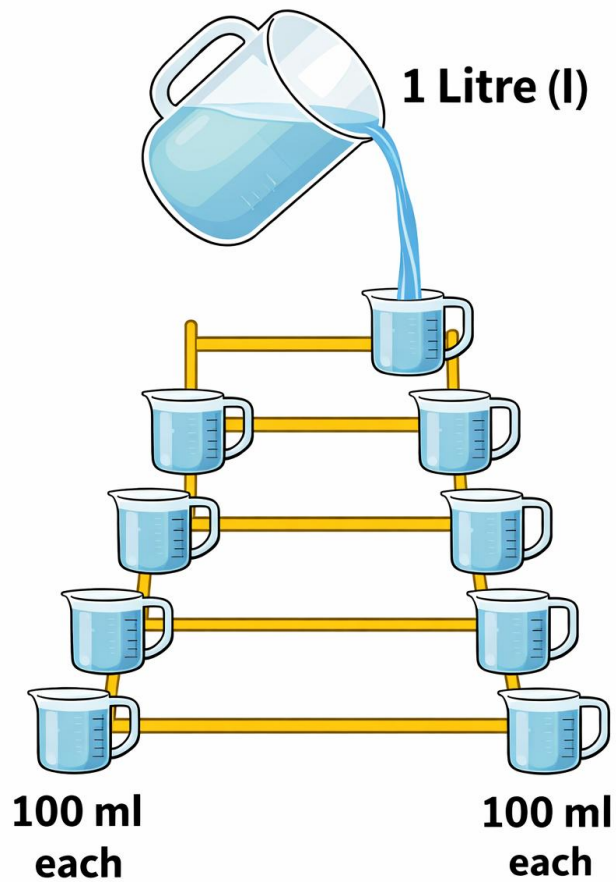
2. Units of Capacity

- **Millilitre (ml):** For very small amounts (medicine dose, teaspoon).
- **Centilitre (cl):** Common on drink bottles (e.g., a 33cl Coke).
- **Litre (l):** For larger amounts (bottles of water, gallons of fuel, buckets).
- **Key Relationship: 1000 millilitres (ml) = 1 litre (l).**

3. Conversion

The conversion is similar to that of grams and kilograms.

- **Litres (l) → Millilitres (ml):** Multiply by 1000.
 - Example: 2.5 litres = $2.5 \times 1000 = 2500$ ml.
- **Millilitres (ml) → Litres (l):** Divide by 1000.
 - Example: 800 ml = $800 \div 1000 = 0.8$ l or $\frac{8}{10}$ l.



$$1\text{ l} = 1000\text{ ml}$$

4. Word Problems

These involve the capacity of everyday containers.

- **Example:** A bucket has a capacity of 12 litres. A jug has a capacity of 3 litres.
How many jugs of water are needed to fill the bucket?
 - Solution: $12\text{ l} \div 3\text{ l/jug} = 4\text{ jugs}$.

EVALUATION

1. Which unit (ml or l) is best for measuring the capacity of: a) a car's fuel tank? b) an eye drop bottle?
2. Convert 3.75 litres to millilitres.
3. Convert 4500 ml to litres.
4. A water dispenser has 5 litres of water. People drink 2800 ml from it. How many millilitres are left?
5. A recipe needs 500 ml of milk. How many times can you make it with a 2-litre pack of milk?

ASSIGNMENT

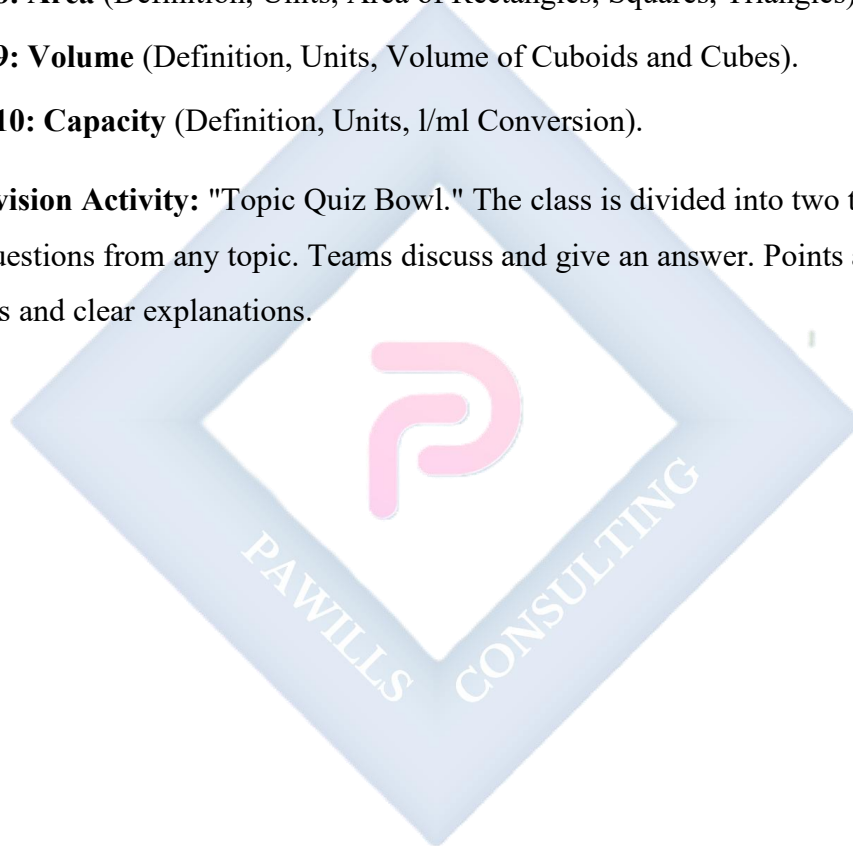
1. **(Written)** A jerrycan holds 25 litres of petrol. How many millilitres is this?
2. **(Practical)** Look at three different liquid containers in your home (e.g., syrup bottle, oil bottle, water bottle). Write down their capacities as shown on the label. Convert one from ml to l, and one from l to ml.
3. **(Oral)** Explain to someone the difference between the *volume* of a bottle and its *capacity*. Use an example.

WEEK 11: REVISION

Comprehensive review of all Second Term topics:

- **Weeks 1-2: Fractions** (Types, Equivalent, Conversions, Addition & Subtraction).
- **Weeks 3-4: Multiplication & Division** (Multi-digit operations, Properties, Remainders, Checking).
- **Week 7: Temperature** (Measurement, °C, Reading Thermometers, Changes).
- **Week 8: Area** (Definition, Units, Area of Rectangles, Squares, Triangles).
- **Week 9: Volume** (Definition, Units, Volume of Cuboids and Cubes).
- **Week 10: Capacity** (Definition, Units, l/ml Conversion).

Suggested Revision Activity: "Topic Quiz Bowl." The class is divided into two teams. The teacher asks questions from any topic. Teams discuss and give an answer. Points are awarded for correct answers and clear explanations.



PRIMARY 5 MATHEMATICS SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Multiplication II (Numbers & Numeration)	Multiplication of 3-digit by 2-digit numbers. Multiplication of 4-digit by 1-digit numbers. Multiplication involving money and measurements. Multiple-step word problems involving multiplication.
2	Division II (Numbers & Numeration)	Division of 3-digit and 4-digit numbers by 2-digit numbers. Long division method. Division involving money and measurements. Multiple-step word problems involving division. Relationship between multiplication and division.
3	Fractions III (Numbers & Numeration)	Multiplication of fractions by whole numbers. Division of fractions by whole numbers (simple cases). Fractions of quantities (finding $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ of amounts). Real-life applications of fractions.
4	Open Sentences (Algebraic Processes)	Introduction to open sentences and unknowns. Using letters to represent unknown numbers. Simple equations: finding the value of unknowns (e.g., $n + 5 = 12$, $3 \times a = 15$). Number patterns and sequences using algebra. Word problems leading to open sentences.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Circle (Mensuration & Geometry)	Parts of a circle (center, radius, diameter, circumference, arc, chord, sector, segment). Relationship between radius and diameter ($d = 2r$). Drawing circles using compasses. Introduction to π (pi) and finding circumference. Practical applications of circles.
8	3-Dimensional Shapes (Mensuration & Geometry)	Properties of 3D shapes (cube, cuboid, cylinder, cone, sphere, pyramid, prism). Identifying faces, edges, and

	Geometry)	vertices. Drawing and constructing 3D shapes. Nets of 3D shapes. Real-life examples of 3D shapes.
9	Structure of Earth (Mensuration & Geometry)	Understanding Earth's spherical shape. Introduction to latitude and longitude. Cardinal points (North, South, East, West) and intermediate directions. Using simple maps and compass directions. Distance and location on Earth's surface.
10	Integrated Problem Solving (All Themes)	Mixed word problems combining multiple operations. Money problems with multiple steps. Measurement problems involving different units. Real-life mathematical investigations. Mathematical games and puzzles.
11	REVISION	Comprehensive review of all topics covered throughout the year
12	EXAMINATION	End of Third Term/Annual Examination
13	CLOSING	Report distribution and long vacation

PRIMARY 5 MATHEMATICS LESSON NOTES

THIRD TERM

WEEK 1: MULTIPLICATION II (NUMBERS & NUMERATION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Multiply 3-digit numbers by 2-digit numbers correctly.
2. Multiply 4-digit numbers by 1-digit numbers.
3. Solve multiplication problems involving money.
4. Solve multiple-step word problems involving multiplication.

CONTENT

1. Multiplying 3-Digit by 2-Digit Numbers

We use the **long multiplication** method. It's like doing two smaller multiplications and adding the results. Remember: when you multiply by the tens digit, you are really multiplying by tens, so you **start your answer in the tens column** (often shown by placing a zero first).

- **Example:** Multiply 123 by 14.
 - **Step 1:** Multiply 123 by the **units** of 14 (4).
 - $4 \times 3 = 12$. Write 2, carry 1 ten.
 - $4 \times 2 = 8$, plus the carried 1 = 9.
 - $4 \times 1 = 4$.
 - First partial product: **492**.
 - **Step 2:** Multiply 123 by the **tens** of 14 (1, which means 10). Write a **0** in the units column first to show you are multiplying by 10.
 - $1 \times 3 = 3$ (write it in the tens column, next to the 0).
 - $1 \times 2 = 2$.

- $1 \times 1 = 1$.
- Second partial product: **1230**.
- **Step 3:** Add the two partial products together: $492 + 1230 = 1722$.

$$\begin{array}{r}
 \begin{array}{ccc}
 & 1 & 2 & 2 \\
 & \xrightarrow{\quad} & \xrightarrow{\quad} & \\
 1 & 2 & 3 & \\
 \times & 1 & 4 & \\
 \hline
 4 & 9 & 2 & \\
 1 & 2 & 3 & 0 \\
 \hline
 1 & 7 & 2 & 2
 \end{array}
 \end{array}$$

Carried Numbers

Placeholder 0

$$1\text{ l} = 1000\text{ ml}$$

2. Multiplying 4-Digit by 1-Digit Numbers

The process is the same as with 3-digit numbers, but you have one more column. Be extra careful with carrying over.

- **Example:** $2,413 \times 3$.
 - 3×3 (units) = 9.
 - 3×1 (tens) = 3.
 - 3×4 (hundreds) = 12. Write 2, carry 1 thousand.
 - 3×2 (thousands) = 6, plus the carried 1 = 7.
 - Answer: **7,239**.

3. Multiplication Involving Money

When multiplying money, ignore the Naira sign (₦) and decimal point at first. Multiply as if they are whole numbers. After getting your answer, put back the decimal point (which separates Naira from Kobo) and the ₦ sign. Remember, two digits come after the decimal point for Kobo.

- **Example:** If one book costs ₦250.00, how much do 5 books cost?
 - Multiply: $250 \times 5 = 1250$.
 - Place the decimal and currency: **₦1,250.00**.

4. Multiple-Step Word Problems

These problems have more than one part. Read carefully, find the first thing you need to calculate, and then use that answer to find the final solution.

- **Example:** A carton contains 12 tins of milk. A truck carries 100 cartons. If one tin costs ₦50, what is the total cost of all the milk?
 - **Step 1:** Find the total number of tins. Tins = $100 \text{ cartons} \times 12 \text{ tins/carton} = 1,200 \text{ tins}$.
 - **Step 2:** Find the total cost. Cost = $1,200 \text{ tins} \times ₦50/\text{tin} = ₦60,000$.

EVALUATION

1. Calculate: 312×23 .
2. Multiply: $4,506 \times 7$.
3. If a school uniform costs ₦3,250, how much will 8 uniforms cost?
4. A factory makes 235 toys every day. How many toys will it make in 30 days?
5. A pack contains 24 pencils. A shop orders 15 packs. How many pencils did they order in total?

ASSIGNMENT

1. **(Written)** Find the product of 708 and 34.
2. **(Practical/Oral)** Go to a shop or market with an adult (or pretend). Choose one item with a known price. Calculate the cost of buying 25 of that item.

3. **(Written)** A printing company binds 1,850 notebooks in one hour. How many notebooks can they bind in an 8-hour workday?



WEEK 2: DIVISION II (NUMBERS & NUMERATION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

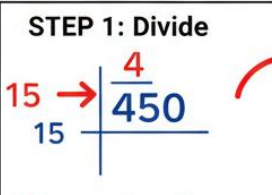
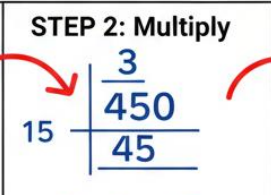
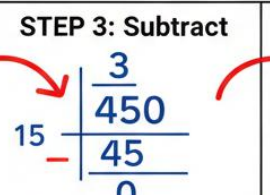
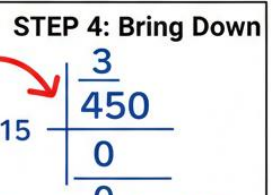
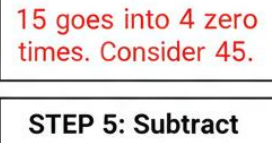
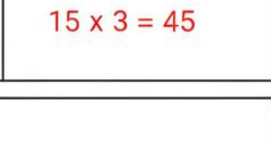
1. Divide 3-digit and 4-digit numbers by 2-digit numbers using long division.
2. Perform division involving money.
3. Solve multiple-step word problems involving division.
4. Explain the relationship between multiplication and division.

CONTENT

1. Long Division with 2-Digit Divisors

The steps (Divide, Multiply, Subtract, Bring Down) are the same, but you must estimate more carefully.

- **Example:** $450 \div 15$.
 - **Step 1:** Can 15 go into 4? No. Can 15 go into 45? Yes. Estimate: $15 \times 3 = 45$. Write 3 above the 5.
 - **Step 2:** Multiply: $3 \times 15 = 45$. Write it under the 45.
 - **Step 3:** Subtract: $45 - 45 = 0$.
 - **Step 4:** Bring down the next digit, which is 0, making 00.
 - **Step 5:** 15 goes into 0 zero times. Write 0 above the 0.
 - **Answer:** **30**.

STEP 1: Divide  15 goes into 4 zero times. Consider 45.	STEP 2: Multiply  $15 \times 3 = 45$	STEP 3: Subtract  $45 - 45 = 0$	STEP 4: Bring Down  Bring down the 0. The number is 00.
STEP 5: Subtract  DMSB	STEP 4: Bring Down  Answer: 30		

2. Division Involving Money

Treat money like any other number during the calculation. At the end, remember to put back the Naira sign and decimal point.

- **Example:** Share ₦500.00 equally among 4 children.
 - Divide: $500 \div 4 = 125$.
 - Place the currency: **₦125.00** per child.

3. Multiple-Step Word Problems

Combine division with another operation (like addition) to solve real-life sharing problems.

- **Example:** 5 friends earned ₦2,000 from a job. They found an extra ₦500. If they share all the money equally, how much does each get?
 - **Step 1:** Find total money. $\text{₦}2,000 + \text{₦}500 = \text{₦}2,500$.
 - **Step 2:** Share equally. $\text{₦}2,500 \div 5 \text{ friends} = \text{₦}500 \text{ each}$.

4. Relationship Between Multiplication and Division

Division is the **exact opposite (inverse)** of multiplication. They undo each other. This is the best way to check your division!

- Fact Family: If $6 \times 8 = 48$, then:
 - $48 \div 8 = 6$
 - $48 \div 6 = 8$
- **Check your work:** (Quotient $\times$ Divisor) + Remainder = Dividend.

EVALUATION

1. Divide: $672 \div 21$.
2. Share ₦1,845 equally among 9 people.
3. Solve: $1,440 \div 24$.
4. If $12 \times 11 = 132$, what is $132 \div 12$?
5. A farmer has 360 eggs. He packs them in trays that hold 18 eggs each. How many trays does he fill?
6. Check your answer for question 1 using multiplication.

ASSIGNMENT

1. **(Written)** Calculate: a) $1,080 \div 15$ b) $2,736 \div 12$.
2. **(Oral/Practical)** With 24 counters (seeds, pebbles), share them equally into groups of 5. How many full groups do you have? How many are left over? Write the division sentence with a remainder.
3. **(Written)** A school needs to transport 300 pupils. If each bus can carry 24 pupils, how many buses are needed? (Hint: Think about what to do with the remainder).

WEEK 3: FRACTIONS III (NUMBERS & NUMERATION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Multiply a fraction by a whole number.
2. Divide a fraction by a whole number.
3. Calculate fractions of quantities (e.g., $\frac{1}{2}$ of 20).
4. Solve real-life problems involving fractions.

CONTENT

1. Multiplying Fractions by Whole Numbers

To multiply a fraction by a whole number, multiply the **numerator** (top number) by the whole number. The denominator stays the same.

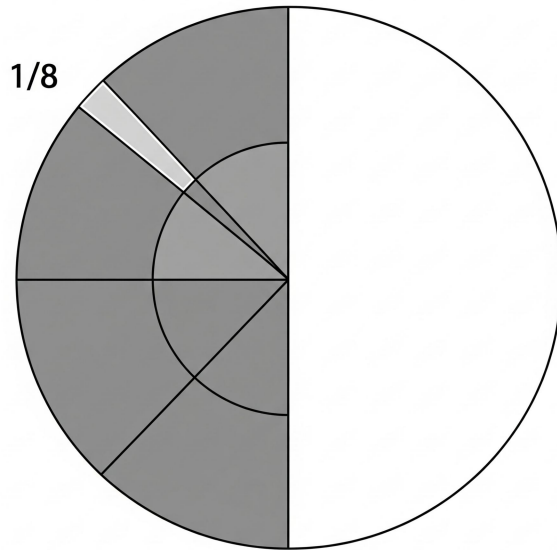
- **Example:** $\frac{2}{7} \times 3$.
 - Multiply the numerator: $2 \times 3 = 6$.
 - Keep the denominator: 7.
 - Answer: $\frac{6}{7}$.
- **Think of it as repeated addition:** $\frac{2}{7} + \frac{2}{7} + \frac{2}{7} = \frac{6}{7}$.

2. Dividing Fractions by Whole Numbers

To divide a fraction by a whole number, multiply the **denominator** (bottom number) by the whole number. The numerator stays the same. You are splitting the fraction into even smaller parts.

- **Example:** $\frac{1}{2} \div 4$.
 - Multiply the denominator: $2 \times 4 = 8$.
 - Keep the numerator: 1.
 - Answer: $\frac{1}{8}$.

- **Think of it:** You have half a pizza. If you share it equally among 4 people, each gets a very small piece, which is $\frac{1}{8}$ of the whole pizza.



3. Fractions of Quantities (Finding "a Fraction Of" a Number)

The word "**of**" in maths often means "**multiply**." Finding a fraction *of* a quantity is the same as multiplying the fraction by the quantity.

- **Example 1:** Find $\frac{1}{4}$ of 20.
 - This means $\frac{1}{4} \times 20$.
 - One method: Divide the whole number by the denominator: $20 \div 4 = 5$.
- **Example 2:** Find $\frac{2}{3}$ of 15.
 - Step 1: Find $\frac{1}{3}$ of 15 first (divide by denominator): $15 \div 3 = 5$.
 - Step 2: Multiply by the numerator: $5 \times 2 = 10$.

4. Real-Life Applications

Fractions are everywhere! We use them for sharing, measuring, and describing parts of a group.

- **Example:** In a class of 40 pupils, $\frac{3}{8}$ are girls. How many girls are in the class?
 - Find $\frac{3}{8}$ of 40.

- $\frac{1}{8}$ of 40 = $40 \div 8 = 5$.
- $\frac{3}{8}$ of 40 = $5 \times 3 = \mathbf{15}$ girls.

EVALUATION

1. Calculate: $\frac{3}{5} \times 6$.
2. Calculate: $\frac{3}{4} \div 2$.
3. What is $\frac{1}{2}$ of 64?
4. Find $\frac{4}{5}$ of 25.
5. A baker uses $\frac{2}{3}$ of a 12kg bag of flour. How many kilograms of flour did he use?

ASSIGNMENT

1. **(Written)** Solve: a) $\frac{5}{9} \times 4$ b) $\frac{4}{5} \div 5$.
2. **(Practical)** At home, find a packet of biscuits or sweets. If it contains 20 items, calculate what $\frac{1}{4}$ of the packet would be. Count out that many items to check.
3. **(Oral)** Create a word problem for a friend where they have to find " $\frac{3}{4}$ of 24." Then, solve it together.

WEEK 4: OPEN SENTENCES (ALGEBRAIC PROCESSES)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify open sentences and unknowns.
2. Use letters (like x , n , a) to represent unknown numbers.
3. Find the value of unknowns in simple equations involving addition, subtraction, and multiplication.
4. Solve word problems leading to open sentences.

CONTENT

1. Introduction to Open Sentences

An **open sentence** is a mathematical statement that is neither completely true nor false because it has a **missing number**. We use a symbol like a box ($\square$), a question mark (?), or most commonly, a **letter** to stand for the unknown.

- Example: $5 + \square = 8$. The missing number is 3.

2. Using Letters for Unknowns

Letters like **x**, **n**, **a**, **b** are used to represent numbers we don't know yet. They are like secret codes we need to crack.

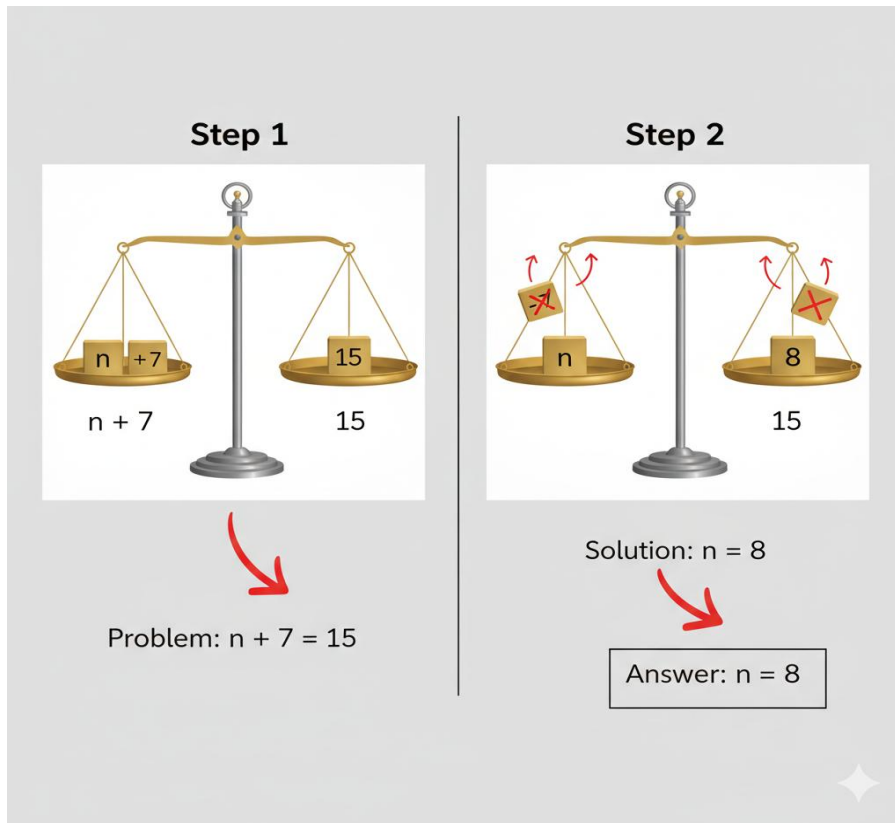
- Example: $n + 5 = 12$. Here, 'n' is the unknown.

3. Solving Simple Equations (Finding the Unknown)

To find the unknown, we need to **isolate it** (get it alone on one side of the $=$ sign). We do the **opposite (inverse) operation**.

- **For Addition ($n + 7 = 15$):** The opposite of adding 7 is **subtracting 7**. So, $n = 15 - 7 = 8$.
- **For Subtraction ($x - 4 = 10$):** The opposite of subtracting 4 is **adding 4**. So, $x = 10 + 4 = 14$.

- **For Multiplication ($3 \times a = 18$):** The opposite of multiplying by 3 is **dividing by 3**. So, $a = 18 \div 3 = 6$.



4. Word Problems Leading to Open Sentences

Read the problem, decide what the unknown number is, and write an equation.

- **Example:** When I add 12 to a number, I get 29. What is the number?
 - Let the number be y .
 - Equation: $y + 12 = 29$.
 - Solve: $y = 29 - 12 = 17$.

EVALUATION

1. Find x : $x + 15 = 32$.
2. Find a : $a - 8 = 20$.
3. Find n : $4 \times n = 36$.
4. Find y : $30 \div y = 5$.

5. I think of a number, multiply it by 6, and get 54. What is the number? (First write the equation).

ASSIGNMENT

1. **(Written)** Solve for the unknown: a) $m - 22 = 40$ b) $5 \times k = 60$.
2. **(Oral/Practical)** Think of a simple riddle like "I add 9 to my secret number and get 21." Tell the riddle to a family member and help them write and solve the open sentence.
3. **(Written)** Ada has some pencils. She gives 7 to her brother and has 11 left. How many pencils did she have at first? Write an open sentence and solve it.



WEEK 5: MIDTERM EXAMINATION

(Assessment covering Weeks 1-4 of Third Term)



WEEK 6: MIDTERM BREAK

(Holiday/Rest Period)



WEEK 7: CIRCLE (MENSURATION & GEOMETRY)

OBJECTIVES

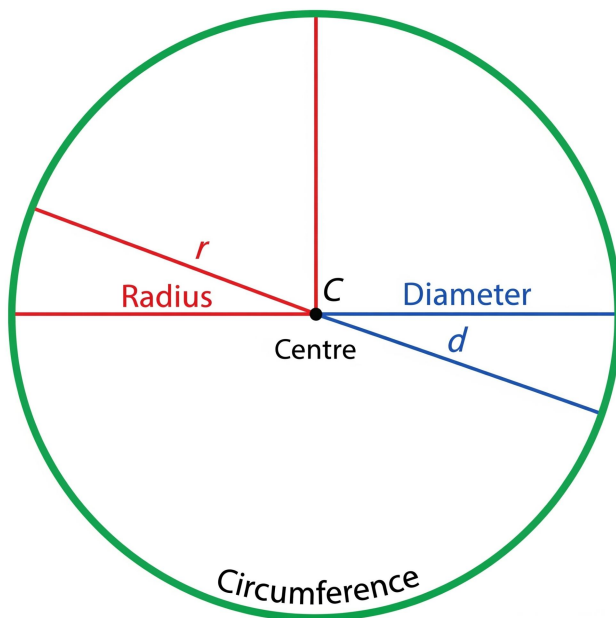
By the end of the lesson, pupils should be able to:

1. Identify and label the parts of a circle (center, radius, diameter, circumference).
2. State the relationship between radius and diameter.
3. Draw a circle using a pair of compasses.
4. Calculate the diameter if given the radius, and vice versa.

CONTENT

1. Parts of a Circle

- **Centre:** The exact middle point of the circle. It is fixed when drawing.
- **Circumference:** The distance all the way **around** the outside edge of the circle.
- **Radius (r):** Any straight line from the **centre** to the **circumference**. All radii (plural) in the same circle are equal.
- **Diameter (d):** A straight line from one side of the circle to the other, passing through the **centre**. It is the longest possible chord.



2. Relationship Between Radius and Diameter

The **diameter** is always twice as long as the **radius**. The **radius** is always half of the **diameter**.

- **Formula: Diameter (d) = $2 \times$ Radius (r).**
- **Formula: Radius (r) = Diameter (d) $\div$ 2.**
- **Example:** If the radius is 7cm, the diameter is $2 \times 7\text{cm} = \mathbf{14\text{cm}}$. If the diameter is 20cm, the radius is $20\text{cm} \div 2 = \mathbf{10\text{cm}}$.

3. Drawing Circles with Compasses

A pair of compasses is the perfect tool for drawing circles.

- **Step 1:** Hold the compass by the top. Place the sharp metal point on your paper where you want the **centre**.
- **Step 2:** Adjust the compass width to match the length of your desired **radius** (use a ruler).
- **Step 3:** Gently rotate the compass 360 degrees around the centre point, keeping the metal point fixed.

4. Introduction to Pi (π)

Pi (π) is a special, constant number that connects the **circumference** of any circle to its **diameter**. For all circles: **Circumference $\div$ Diameter = π .**

- π is approximately **3.14** or $\frac{22}{7}$.
- We will use this more in future lessons to calculate circumference.

EVALUATION

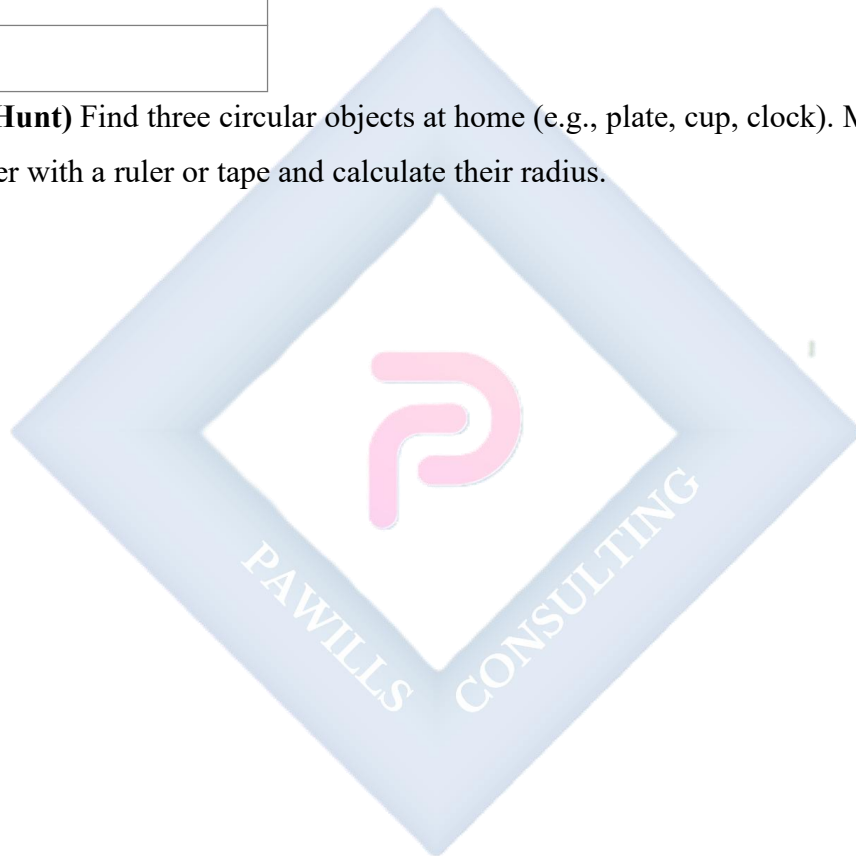
1. In a circle, what is the name for a line from the centre to the edge?
2. If the diameter of a wheel is 56cm, what is its radius?
3. Draw a circle with a radius of 4cm. Label the centre, a radius, and the diameter.
4. A circular plate has a radius of 10cm. What is its diameter?
5. What is the approximate value of π ?

ASSIGNMENT

1. **(Practical/Drawing)** Use a compass to draw two circles: one with a radius of 3cm and another with a diameter of 8cm.
2. **(Written)** Complete this table:

Radius (r)	Diameter ($d = 2 \times r$)
6 cm	
	30 m
12.5 mm	

3. **(Oral/Hunt)** Find three circular objects at home (e.g., plate, cup, clock). Measure their diameter with a ruler or tape and calculate their radius.



WEEK 8: 3-DIMENSIONAL SHAPES (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify common 3D shapes (cube, cuboid, cylinder, cone, sphere).
2. List the properties of 3D shapes (faces, edges, vertices).
3. Differentiate between a prism and a pyramid.
4. Identify real-life objects that correspond to 3D shapes.

CONTENT

1. Properties of 3D Shapes

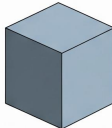
3D (three-dimensional) shapes are **solid** objects. They have length, width/breadth, and height. We describe them by:

- **Faces:** The flat or curved surfaces.
- **Edges:** The lines where two faces meet.
- **Vertices (singular: vertex):** The corners or points where edges meet.

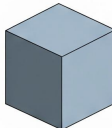
2. Common 3D Shapes and Their Properties

- **Cube:** (e.g., Dice, Sugar cube)
 - Faces: 6 (all are squares)
 - Edges: 12
 - Vertices: 8
- **Cuboid (Rectangular Prism):** (e.g., Matchbox, Book)
 - Faces: 6 (all are rectangles)
 - Edges: 12
 - Vertices: 8
- **Cylinder:** (e.g., Tin of milk, Pipe)
 - Faces: 2 flat circular faces, 1 curved face.

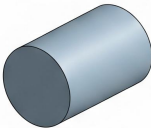
- Edges: 2 circular edges.
- Vertices: 0
- **Cone:** (e.g., Traffic cone, Ice-cream cone)
 - Faces: 1 flat circular face, 1 curved face.
 - Edges: 1 circular edge.
 - Vertices: 1 point (apex)
- **Sphere:** (e.g., Ball, Globe, Orange)
 - Faces: 1 curved face.
 - Edges: 0
 - Vertices: 0



Faces	Edges	Vertices
Cube	6	6



Faces	Edges	Vertices
Cuboid	12	8



Cylinder	Edges	Vertices
Vertices	24	4



Cone	Edges	Vertices
Sylinder	120	24



Facere	Edges	Vertices
Sphere	25	8

3. Prisms vs. Pyramids

- **Prism:** A shape with the **same cross-section** (same shape) all the way through. It has two identical ends. Examples: Cuboid (rectangular prism), Triangular Prism (Toblerone box).

- **Pyramid:** A shape with a **polygon base** and **triangular faces** that meet at a single point (apex). Example: Egyptian pyramid (square pyramid).
4. **Nets of 3D Shapes** A **net** is a flat, 2D pattern that can be folded to make a 3D shape. Imagine carefully opening up a cardboard box and laying it flat—that's its net.

EVALUATION

1. How many vertices does a cube have?
2. Name a real-life object that is a cylinder.
3. What is the main difference between a cube and a cuboid?
4. Which shape has 1 vertex and 1 circular face?
5. Is a triangular prism a pyramid? Why or why not?

ASSIGNMENT

1. **(Written)** Make a table. List Cube, Cuboid, Cylinder, Cone, Sphere. For each, write the number of faces, edges, and vertices.
2. **(Drawing/Practical)** Find a small box (cuboid) at home. Carefully take it apart or imagine unfolding it. Draw what its net would look like.
3. **(Oral/Hunt)** Go on a "3D Shape Safari" in your home. Find and name one example of a cube, a cuboid, a cylinder, and a sphere.

WEEK 9: STRUCTURE OF EARTH (MENSURATION & GEOMETRY)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Describe the shape of the Earth (Sphere).
2. Identify the 4 Cardinal points (North, South, East, West).
3. Use a simple compass to find directions.
4. Locate simple positions using a grid map.

CONTENT

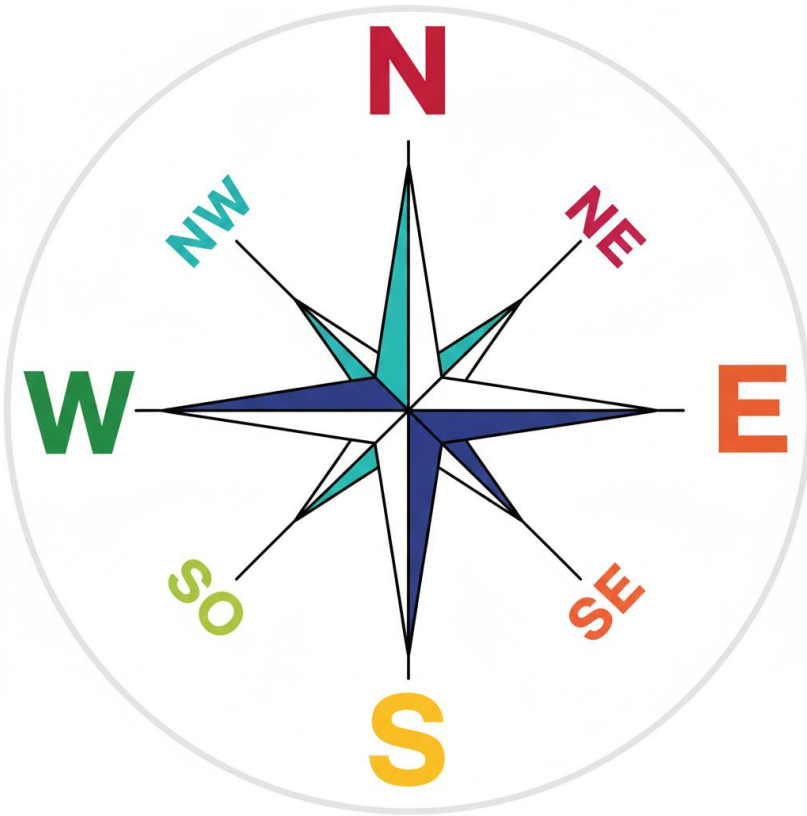
1. Shape of the Earth

The Earth is not flat like a plate. It is a **sphere** (a 3D shape like a ball). It is slightly flattened at the poles, but we generally call it a sphere. It spins around an imaginary line called its **axis**, which goes from the North Pole to the South Pole.

2. Cardinal Points (Directions)

These are the four main directions we use to navigate.

- **North (N):** The direction towards the North Pole. On most maps, it points **up**.
- **South (S):** The direction towards the South Pole. Opposite of North.
- **East (E):** The direction where the sun **rises** in the morning.
- **West (W):** The direction where the sun **sets** in the evening.
- **In-Between Points:** North-East (NE), North-West (NW), South-East (SE), South-West (SW).



3. Using a Compass

A magnetic compass has a needle that always points to **Magnetic North**. You can use it to find all other directions. To use it:

- Hold the compass flat and steady.
- Wait for the needle to stop moving.
- Turn the compass so the "N" on its dial aligns with the red part of the needle.
- Now you can read all the other directions.

4. Introduction to Simple Grid Maps

A map is a flat drawing of a place. A **grid map** uses letters and numbers (like in a game of Battleship) to help locate places easily.

- The vertical columns are often labeled with **letters** (A, B, C...).
- The horizontal rows are often labeled with **numbers** (1, 2, 3...).

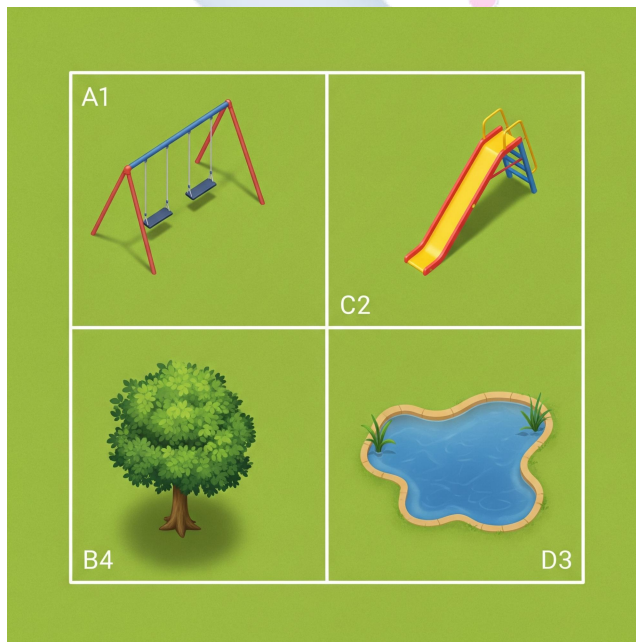
- A location is given as a **grid reference**, like **B3**. Find column B, then go across to row 3.

EVALUATION

1. What is the shape of the Earth?
2. Which cardinal direction is opposite to East?
3. If you are facing North and turn 90 degrees to your right, which direction are you facing now?
4. What does a compass needle always point to?
5. On a grid map, what is found at grid reference **D5**?

ASSIGNMENT

1. **(Drawing)** Draw your own Compass Rose. Colour North red, South blue, East green, and West yellow.
2. **(Practical)** On a sunny morning, go outside. Note where the sun is. That is roughly East. Now, determine where North, South, and West are.
3. **(Written)** Here is a simple grid map of a park.



Answer: a) Where is the slide? b) What is in square B4?

WEEK 10: INTEGRATED PROBLEM SOLVING (ALL THEMES)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Solve mixed word problems involving multiple operations (+, -, x, ÷).
2. Solve real-life problems involving money, length, and time together.
3. Apply logical reasoning to solve mathematical puzzles.

CONTENT

1. Mixed Operation Problems

Real-life problems often need more than one step. Read slowly. Decide what you need to find first. Remember the **order of operations** (BODMAS/BIDMAS) if needed: solve Brackets first, then Of/Orders (powers), then Division/Multiplication (left to right), then Addition/Subtraction (left to right).

- **Example:** A trader bought 50 bags of rice. He sold 20 on Monday and 15 on Tuesday. How many bags are left?
 - Total sold = $20 + 15 = 35$ bags.
 - Left = $50 - 35 = 15$ bags.

2. Combining Money and Measurement

These problems require you to use formulas (like perimeter) and then apply a cost.

- **Example:** Mr. Ahmed wants to fence his square garden. One side is 12m. Fencing costs ₦500 per metre. What is the total cost?
 - **Step 1 (Find Perimeter):** Perimeter of square = $4 \times \text{side} = 4 \times 12\text{m} = 48\text{m}$.
 - **Step 2 (Find Cost):** Total cost = $48\text{m} \times ₦500/\text{m} = ₦24,000$.

3. Combining Time and Distance (Speed Concept)

If you know how far something travels in a set time, you can find how far it goes in a different time.

- **Example:** A cyclist rides 15 km in 1 hour. How far can she ride in 4 hours at the same speed?

- $\text{Distance} = 15 \text{ km/hour} \times 4 \text{ hours} = \mathbf{60 \text{ km.}}$

4. Mathematical Investigations and Puzzles

Use logic and reasoning, not just calculation.

Example:

I am an odd number. I am less than 25 but greater than 18. The sum of my digits is 6. What number am I?

Numbers between 18 and 25 are: 19, 20, 21, 22, 23, 24.

Odd numbers are: 19, 21, 23.

Sum of digits:

- $1 + 9 = 10$
- $2 + 1 = 3$
- $2 + 3 = 5$

None of these has a digit sum of 6, so the puzzle (as written) has **no solution** and needs correction.

A better example (with a valid answer):

I am an odd number between 10 and 20. The sum of my digits is 8. What number am I?

Odd numbers between 10 and 20: 11, 13, 15, 17, 19.

Digit sums:

- $1 + 1 = 2$
- $1 + 3 = 4$
- $1 + 5 = 6$
- $1 + 7 = 8 \checkmark$
- $1 + 9 = 10$

Answer: 17.

EVALUATION

1. A school bought 200 notebooks. They gave 45 to Primary 5A and 38 to Primary 5B. How many notebooks are left?
2. To make a dress, a tailor needs 3 metres of cloth. If cloth costs ₦1,200 per metre, how much will 5 dresses cost?
3. A journey starts at 9:15 am and ends at 1:45 pm. How long does the journey last?
4. A rectangular poster has an area of 60cm^2 . Its length is 10cm. What is its width?
5. (Puzzle) I am a number. If you divide me by 3 and then add 5, you get 10. What number am I?

ASSIGNMENT

1. **(Written)** Musa saves ₦200 every week for 8 weeks. He then spends ₦750 on a pair of shoes. How much money does he have left?
2. **(Practical)** Measure the perimeter of your bedroom or sitting room. If carpet costs ₦800 per metre to install along the edge, estimate the total cost. (You can use estimated measurements).
3. **(Oral/Puzzle)** Create a number puzzle for a classmate to solve. It should involve two steps (like the one in evaluation question 5).

WEEK 11: REVISION

CONTENT

Comprehensive review of all Third Term topics:

- **Week 1: Multiplication II** (3-digit by 2-digit, 4-digit by 1-digit, Money, Multi-step).
- **Week 2: Division II** (Long division by 2-digit, Money, Multi-step, Inverse Check).
- **Week 3: Fractions III** (Multiplying/Dividing fractions by whole numbers, Fractions of quantities).
- **Week 4: Open Sentences** (Identifying unknowns, Solving simple equations, Word problems).
- **Week 7: Circle** (Parts, Radius-Diameter relationship, Drawing, Pi).
- **Week 8: 3D Shapes** (Properties, Prisms vs. Pyramids, Real-life examples).
- **Week 9: Structure of Earth** (Shape, Cardinal Points, Using a compass, Grid maps).
- **Week 10: Integrated Problem Solving** (Mixed operations, Combined themes, Puzzles).

Suggested Revision Activity: "Concept Poster Gallery Walk." In groups, pupils create a simple poster summarizing one topic with key points and examples. Posters are displayed, and pupils walk around to quiz each other.